# Dorey's Resource Book

## Topic 1 – Differential Calculus

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# 1 The derivative of $e^{f(x)}$

ANCHOR QUESTION

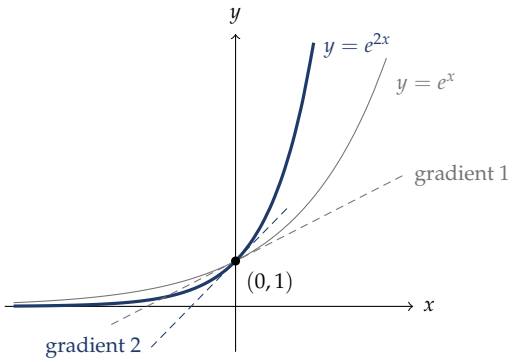
*Differentiate* $y = e^{x^2-5x}$.

SYLLABUS MAV-12-04

- Apply the rules of differentiation to find the derivative of $f(x) = ke^{ax}$, where $k$ and $a$ are constants
- Use the chain rule to prove $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where $a$ and $b$ are constants and $a \neq 0$
- Use the chain rule to differentiate functions of the form $e^{f(x)}$

NOTES

- $\frac{d}{dx}\left(e^x\right) = e^x$.
- $\frac{d}{dx}\left(e^{f(x)}\right) = f'(x)\,e^{f(x)}$. This is on the reference sheet.
- The power stays exactly the same. Multiply by the derivative of the power.
- Special case: $\frac{d}{dx}\left(e^{ax+b}\right) = a\,e^{ax+b}$, and $\frac{d}{dx}\left(ke^{ax}\right) = ka\,e^{ax}$.
- Proof: let $u = ax + b$, so $y = e^u$. Then $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.
- $x$ in the power of $e$: use this rule. $x$ in the base, as in $x^e$: use the power rule. No $x$, as in $e^4$: a constant, derivative 0.
- Rewrite first with index laws: $\frac{1}{e^{kx}} = e^{-kx}$, $e^a e^b = e^{a+b}$, $\sqrt{e^x} = e^{x/2}$.
- $e^{f(x)} > 0$ for every $x$, so $f'(x)\,e^{f(x)} = 0$ only when $f'(x) = 0$.




WORKED EXAMPLE

*Differentiate* $y = e^{3x^2-x}$.

$f(x) = 3x^2 - x$, so $f'(x) = 6x - 1$. the power and its derivative

$$\frac{dy}{dx} = f'(x)\,e^{f(x)}$$
$$= (6x - 1)\,e^{3x^2-x}$$

## Tier 1 · the derivative of $e^{f(x)}$

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**1.** For each function, write the power $f(x)$, then find $f'(x)$.

a) $e^{9x}$

b) $e^{2-x}$

c) $e^{x^4-3x}$

**2.** Differentiate:

a) $y = e^{7x}$

b) $y = e^{x^2+1}$

c) $y = e^{x^3-6x}$

**3.** Differentiate:

a) $y = e^{1-5x}$

b) $y = e^{\sqrt{x}}$

c) $y = e^4$

## Tier 2 · a coefficient in front

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**4.** Differentiate:

a) $y = 4e^{3x}$

b) $y = -2e^{5x+1}$

c) $y = \frac{1}{2}e^{6x}$

**5.** Differentiate:

a) $y = 3e^{x^2}$

b) $y = -e^{2x^2-x}$

**6.** Show that $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = 2e^{8x-2}$.

## Tier 3 · sums and differences

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**7.** Differentiate:

a) $y = e^{4x} + e^{-4x}$

b) $y = 3e^{2x} - x^3$

**8.** Differentiate $y = 2e^{-3x} - 5e^{x} + 7x$.

**9.** Differentiate $y = x^2 - 4x + e^{2-x}$.

## Tier 4 · rewrite with index laws first

**10.** Differentiate:

a) $y = \dfrac{1}{e^x}$

b) $y = \dfrac{4}{e^{2x}}$

**11.** Differentiate:

a) $y = e^x \cdot e^{3x}$

b) $y = \left(e^{2x}\right)^3$

**12.** Differentiate:

a) $y = \sqrt{e^{4x}}$

b) $y = (e^x + e^{-x})^2$, by expanding first

## Tier 5 · exact values of the derivative

**13.** If $f(x) = e^{2x+1}$, find the exact value of $f'(0)$.

**14.** If $f(x) = e^{x^2-4}$, find the exact value of $f'(2)$.

**15.** Find the gradient of the tangent to $y = 3e^{-x} + e^x$ at the point where $x = \ln 3$.

## Tier 6 · the chain rule on the outside

**16.** Differentiate $y = (e^x + 1)^3$.

**17.** Differentiate $y = \left(2 - e^{3x}\right)^4$.

**18.** Differentiate $y = \sqrt{1 + e^{2x}}$.

## Tier 7 · show that, then hence

**19.** Let $y = e^{x^2} \cdot e^{-2x}$.

a) Show that $\dfrac{dy}{dx} = 2(x-1)\,e^{x^2-2x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

**20.** Let $y = \dfrac{e^{2x^2}}{e^{8x}}$.

a) Show that $\dfrac{dy}{dx} = 4(x-2)\,e^{2x^2-8x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

**21.** Let $f(x) = e^{x^3-3x}$.

a) Find $f'(x)$ in fully factorised form.

b) Find the exact coordinates of the points where the tangent is horizontal.

## Conceptual

**22.** A student writes $\frac{d}{dx}\left(e^{3x}\right) = 3x\,e^{3x-1}$. Explain the error and give the correct derivative.

**23.** Use the chain rule to prove that $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where $a$ and $b$ are constants.

**24.** Explain why the graph of $y = 5e^{2x}$ is increasing for every value of $x$.

**25.** The curve $y = e^{x^2+kx}$ has a horizontal tangent at $x = 3$. Find the value of $k$.

**26.** Let $y = 5e^{2x}$.

(a) Find $\frac{dy}{dx}$. (1 mark)

(b) Show that $\frac{dy}{dx} = 2y$. (1 mark)

(c) Hence find the gradient of the curve at the point where $y = 10$. (1 mark)

## Revision

**27.** Differentiate $(x^2+5)^7$ with respect to $x$. (2 marks)

**28.** Find the gradient of the tangent to the curve $y = x^4 - 2x$ at the point $(1, -1)$. (2 marks)

**29.** Let $f(x) = x(2x-1)^4$. Show that $f'(x) = (2x-1)^3(10x-1)$. (2 marks)

**30.** Differentiate $\frac{3x}{x^2+1}$ with respect to $x$. (2 marks)

**31.** Find the equation of the tangent to the curve $y = (2x-1)^3$ at the point where $x = 1$. (3 marks)

**32.** Find the equation of the normal to the curve $y = x^2 - 3x$ at the point $(1, -2)$. Give your answer in general form. (2 marks)

**33.** Let $f(x) = e^x - 1$. Show that $f(x) \times f(-x) + f(x) + f(-x) = 0$. (2 marks)

**34.** Solve $\ln x = 3$. Give your answer correct to four decimal places. (2 marks)

**35.** Solve $2^{2x-1} = 32$. (2 marks)

**36.** Solve $\ln x - \frac{2}{\ln x} = 1$. (3 marks)

**37.** Given that $\log_a 2 = 0.36$ and $\log_a 3 = 0.57$, find $\log_a 12$. (2 marks)

**38.** Write $\ln\left(\frac{x^3}{\sqrt{x+1}}\right)$ in terms of $\ln x$ and $\ln(x+1)$. (2 marks)

**39.** Find the exact value of $\theta$ such that $2\sin\theta = \sqrt{3}$, where $\dfrac{\pi}{2} \leq \theta \leq \pi$. (2 marks)

**40.** Find the exact values of $x$ such that $2\cos x = -\sqrt{2}$, where $0 \leq x \leq 2\pi$. (2 marks)

**41.** Solve $2\sin^2\dfrac{x}{2} = 1$ for $-\pi \leq x \leq \pi$. (3 marks)

**42.** Show that $\dfrac{1-\cos^2\theta}{\sin\theta\cos\theta} = \tan\theta$. (2 marks)

**43.** What is the domain of $f(x) = \dfrac{1}{\sqrt{x+2}}$? (1 mark)

**44.** Given $f(x) = 2x+1$ and $g(x) = x^2 - 3$, find $f(g(x))$ and solve $f(g(x)) = 7$. (2 marks)

**45.** Show that $f(x) = x^3 - 4x$ is an odd function. (1 mark)

**46.** Solve the inequality $x^2 - 2x - 15 < 0$. (2 marks)

**47.** Find the inverse function of $f(x) = 3x - 4$. (1 mark)

**48.** Describe the transformations that map $y = f(x)$ onto $y = 2f(x-1) + 3$. (2 marks)

**49.** The probability that Amy wins a prize in her weekly raffle is $\dfrac{1}{5}$. The probability that Ben wins a prize in his is $\dfrac{1}{4}$. What is the probability that, in a given week, at least one of them wins a prize? (2 marks)

**50.** A box holds 10 chocolates: 4 mint and 6 caramel. Two are chosen at random and eaten.

(a) What is the probability that both are mint? (1 mark)

(b) What is the probability that they have different centres? (1 mark)

**51.** Solve $|2x - 1| \leq 7$. (2 marks)

**52.** Find the angle that the tangent to $y = x^2$ at the point where $x = \frac{1}{2}$ makes with the positive direction of the $x$-axis. (2 marks)

# 2 Derivatives of $a^x$, products and quotients

ANCHOR QUESTION

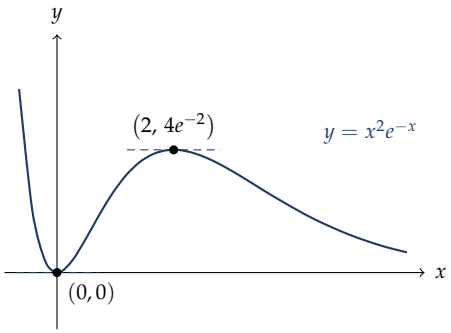
*Differentiate $y = x^2e^{3x}$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Prove and use the formula $\frac{d}{dx}(a^x) = (\ln a)\,a^x$, where $a$ is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course

NOTES

- Any positive number is a power of $e$: $a = e^{\ln a}$, so $a^x = e^{x\ln a}$.
- $\frac{d}{dx}(a^x) = (\ln a)\,a^x$. Proof: $\frac{d}{dx}\left(e^{x\ln a}\right) = \ln a \cdot e^{x\ln a} = (\ln a)\,a^x$.
- $\ln a$ is just a number. When $a = e$, $\ln e = 1$ and the rule gives back $\frac{d}{dx}(e^x) = e^x$.
- Product rule: if $y = uv$, then $\frac{dy}{dx} = u'v + uv'$.
- Quotient rule: if $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$. Both rules are on the reference sheet.
- After the product rule, both terms share the exponential: take it out as a common factor, e.g. $2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$.
- An exponential is never 0, so a factorised derivative is 0 only where its other factors are 0.




WORKED EXAMPLE

*Differentiate $y = x^2e^{5x}$, giving your answer in factorised form.*

$u = x^2,\ u' = 2x; \quad v = e^{5x},\ v' = 5e^{5x}$ — name the parts

$\frac{dy}{dx} = u'v + uv' = 2xe^{5x} + 5x^2e^{5x}$ — product rule

$= xe^{5x}(2 + 5x)$ — common factor $xe^{5x}$

## Tier 1 · the product rule with $e$

**53.** Write as a power of $e$.

a) $9^x$

b) $2^{5x}$

**54.** Factorise fully.

a) $3x^2e^{2x} + 2x^3e^{2x}$

b) $e^{3x} - 3xe^{3x}$

**55.** Differentiate:

a) $y = 6^x$

b) $y = 5 \cdot 2^x$

c) $y = 3^{4x}$

**56.** Differentiate, giving each answer in factorised form.

a) $y = xe^x$

b) $y = x^2e^{4x}$

**57.** Differentiate $y = x^4e^x$, giving your answer in factorised form.

## Tier 2 · a negative power

**58.** Differentiate $y = (3x - 1)e^{-x}$, giving your answer in factorised form.

**59.** Differentiate $y = xe^{-2x}$, giving your answer in factorised form.

**60.** Differentiate $y = x^2e^{-3x}$, giving your answer in factorised form.

## Tier 3 · the quotient rule

**61.** Differentiate $y = \dfrac{e^{3x}}{x}$.

**62.** Differentiate $y = \dfrac{x}{e^{2x}}$, simplifying your answer.

**63.** Differentiate $y = \dfrac{e^x}{x+1}$.

## Tier 4 · products with $a^x$

**64.** Differentiate $y = x \cdot 2^x$.

**65.** Differentiate $y = x^2 \cdot 5^x$, giving your answer in factorised form.

**66.** Differentiate $y = 2^x e^x$.

## Tier 5 · quotients of exponentials

**67.** Differentiate $y = \dfrac{e^x}{e^x + 2}$.

**68.** Differentiate $y = \dfrac{e^x - 1}{e^x + 1}$.

**69.** Differentiate $y = \dfrac{e^{2x}}{1 + e^{2x}}$.

## Tier 6 · horizontal tangents

**70.** Find the exact coordinates of the point on $y = xe^{-x}$ where the tangent is horizontal.

**71.** Find the exact coordinates of the points on $y = x^2 e^x$ where the tangent is horizontal.

**72.** Find the exact coordinates of the point on $y = \dfrac{e^x}{x}$ where the tangent is horizontal.

## Conceptual

**73.** Prove that $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$, where $a > 0$.

**74.** A student writes $\dfrac{d}{dx}\left(2^x\right) = x \cdot 2^{x-1}$. Explain the error and give the correct derivative.

**75.** Show that $\dfrac{d}{dx}\left(xe^x\right) = (x+1)e^x$, and hence find the gradient of $y = xe^x$ at the origin.

**76.** Find the exact gradient of the tangent to $y = xe^{2x}$ at the point where $x = 1$.

**77.** Let $f(x) = xe^{-x}$.

(a) Show that $f'(x) = (1 - x)e^{-x}$. (1 mark)

(b) Explain why $f(x)$ is increasing for $x < 1$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

## Revision

**78.** Differentiate $(e^x + 2)^3$ with respect to $x$. (2 marks)

**79.** Differentiate $5e^{-2x}$ with respect to $x$. (1 mark)

**80.** Let $f(x) = e^{x^2-2x}$. Find the exact value of $f'(2)$. (2 marks)

**81.** Show that $y = 3e^{2x}$ satisfies $\dfrac{dy}{dx} = 2y$. (1 mark)

**82.** The curve $y = e^{x^2+kx}$ has a horizontal tangent at $x = -2$. Find the value of $k$. (2 marks)

**83.** Solve $e^{2x} - 5e^x + 6 = 0$, giving exact answers. (3 marks)

**84.** Solve $3^x = 20$. Give your answer correct to two decimal places. (2 marks)

**85.** Use the change of base rule to evaluate $\log_5 40$, correct to three decimal places. (1 mark)

**86.** Given that $\ln 2 = a$ and $\ln 5 = b$, express $\ln 40$ and $\ln 0.4$ in terms of $a$ and $b$. (2 marks)

**87.** Differentiate $\sqrt{16 - x^2}$ with respect to $x$. (2 marks)

**88.** Differentiate $\dfrac{x+1}{2x-1}$ with respect to $x$. (2 marks)

**89.** Find the equation of the tangent to the curve $y = x^3 - 2x^2 + 1$ at the point $(2,\ 1)$. (2 marks)

**90.** The point $A(-2,\ 4)$ lies on the parabola $y = x^2$. Find the equation of the tangent to the parabola at $A$. (2 marks)

**91.** Find the values of $k$ for which the quadratic equation $x^2 - (k+2)x + (k+5) = 0$ has equal roots. (3 marks)

**92.** Solve $x^2 = 5x$. (1 mark)

**93.** Solve $\sqrt{3}\tan x = 1$ for $0 \leq x \leq 2\pi$. (2 marks)

**94.** Solve $2\cos^2 x = 1$ for $0 \leq x \leq 2\pi$. (3 marks)

**95.** A sector of a circle has radius 6 cm and arc length $4\pi$ cm.

(a) Find the angle of the sector in radians. (1 mark)

(b) Find the exact area of the sector. (1 mark)

**96.** What is the domain of $y = \ln(2x - 6)$? (1 mark)

**97.** Given $f(x) = x^2$ and $g(x) = x + 3$, find $f(g(x))$ and state its range. (2 marks)

**98.** Show that $f(x) = x^4 - 5x^2$ is an even function. (1 mark)

**99.** Solve $|3 - 2x| \leq 9$. (2 marks)

**100.** Kim has three red shirts and two yellow shirts. On Monday and on Tuesday she chooses one shirt at random to wear, and she does not wear the same shirt twice. What is the probability that she wears the same colour on both days? (2 marks)

**101.** Let $M$ be the midpoint of $(-3, 2)$ and $(5, 6)$. Find the equation of the line through $M$ with gradient $-\frac{1}{2}$. Give your answer in general form. (2 marks)

**102.** Find integers $a$ and $b$ such that $\frac{3}{\sqrt{7}-2} = a + b\sqrt{7}$. (2 marks)

# 3 Tangents and normals with exponentials

ANCHOR QUESTION

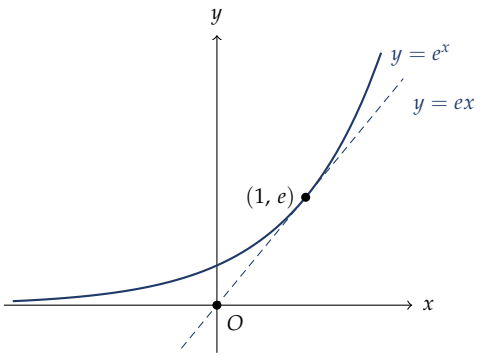
*Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 1$.*

SYLLABUS MAV-12-04

- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- The gradient of the tangent at $x = a$ is $f'(a)$.
- The normal is perpendicular to the tangent, so its gradient is $-\dfrac{1}{f'(a)}$.
- Find the point with the original function: $y = f(a)$. Find the gradient with the derivative.
- Use $y - y_1 = m(x - x_1)$, then rearrange to $y = mx + c$ or general form.
- Leave $e$ in exact answers: $e^0 = 1$, $e^1 = e$, $e^{\ln k} = k$.




WORKED EXAMPLE

*Find the equation of the tangent to $y = e^{4x}$ at the point where $x = 1$.*

$x = 1$: $y = e^4$, so the point is $(1, e^4)$. — point from $f$

$\dfrac{dy}{dx} = 4e^{4x}$, so $m = 4e^4$. — gradient from $f'$

$y - e^4 = 4e^4(x - 1)$ — point–gradient form

$y = 4e^4x - 3e^4$

## Tier 1 · tangents to $y = e^{kx}$

**103.** Find the exact value.

a) $e^{4x}$ at $x = 1$

b) $3e^{1-x}$ at $x = 1$

c) $xe^{3x}$ at $x = 1$

**104.** The tangent has gradient $m$. Find the gradient of the normal.

a) $m = 5e$

b) $m = e^4$

c) $m = -\dfrac{3}{e}$

**105.** Find the equation of the tangent to $y = e^x$ at the point where $x = 2$.

**106.** Find the equation of the tangent to $y = e^{5x}$ at the point where $x = 0$.

**107.** Find the equation of the tangent to $y = 2e^{3x}$ at the point where $x = 1$.

## Tier 2 · normals

**108.** Find the equation of the normal to $y = e^{3x}$ at the point where $x = 0$. Give your answer in general form.

**109.** Find the equation of the normal to $y = e^x$ at the point where $x = 1$. Give your answer in general form.

**110.** Find the equation of the normal to $y = e^{-2x}$ at the point where $x = 0$. Give your answer in general form.

## Tier 3 · tangents that need the product rule

**111.** Find the equation of the tangent to $y = xe^{2x}$ at the origin.

**112.** Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$.

**113.** Find the equation of the tangent to $y = x^2e^x$ at the point where $x = 1$.

## Tier 4 · normals in general form

**114.** Find the equation of the normal to $y = e^{x^2}$ at the point where $x = -1$. Give your answer in general form.

**115.** Find the equation of the normal to $y = e^{2x-2}$ at the point where $x = 1$. Give your answer in general form.

**116.** Find the equation of the normal to $y = e^{x^2-4}$ at the point where $x = 2$. Give your answer in general form.

## Tier 5 · show that

**117.** Show that the tangent to $y = e^{2x}$ at the point where $x = a$ meets the $x$-axis at $x = a - \frac{1}{2}$.

**118.** Show that the tangent to $y = e^{-x}$ at the point where $x = a$ meets the $x$-axis at $x = a + 1$.

**119.** Show that the tangent to $y = e^x$ at the point where $x = a$ meets the $y$-axis at $y = (1 - a)e^a$.

## Tier 6 · finding the point

**120.** The tangent to $y = e^{2x}$ at the point where $x = a$ passes through the origin. Find $a$ and the equation of the tangent.

**121.** Find the exact coordinates of the point on $y = e^x$ where the tangent has gradient 3.

**122.** Find the exact coordinates of the point on $y = e^{3x}$ where the normal has gradient $-\frac{1}{6}$.

## Conceptual

**123.** Explain why no tangent to $y = e^x$ is horizontal.

**124.** Show that the tangents to $y = e^x$ and $y = e^{-x}$ at the point $(0, 1)$ are perpendicular.

**125.** Find where the normal to $y = e^x$ at the point where $x = 0$ crosses the $x$-axis.

**126.** A student finds the tangent to $y = e^{2x}$ at $x = 1$ using the point $(1, 2e^2)$. Explain the error.

**127.** Let $f(x) = e^{2x}$.

(a) Find the equation of the tangent to $y = f(x)$ at the point where $x = 0$. (2 marks)

(b) Find where this tangent meets the $x$-axis. (1 mark)

## Revision

**128.** Find the derivative of $y = x^2e^{3x}$. (2 marks)

**129.** Differentiate $\dfrac{e^{2x}}{x+1}$ with respect to $x$. (2 marks)

**130.** Differentiate $3^{2x}$ with respect to $x$. (1 mark)

**131.** Differentiate $(1-e^x)^2$ with respect to $x$. (2 marks)

**132.** Let $f(x) = xe^{-2x}$. Show that $f'(x) = e^{-2x}(1-2x)$. (2 marks)

**133.** The curve $y = ae^{bx}$ passes through $(0, 4)$, and its gradient at that point is $-8$. Find $a$ and $b$. (2 marks)

**134.** Solve $3^{2x+1} = 81$. (2 marks)

**135.** Solve $2^x = 10$. Give your answer correct to three significant figures. (2 marks)

**136.** Solve $2^{2x} - 3(2^x) - 4 = 0$. (3 marks)

**137.** The line $AB$ has equation $y = 3x - 3$. It meets the $y$-axis at $A$ and the $x$-axis at $B$. The line $BC$ is perpendicular to $AB$, and $C$ lies on the $y$-axis.

(a) Find the equation of the line $BC$. (2 marks)

(b) Find the area of triangle $ABC$. (2 marks)

**138.** Show that the lines $2x + 5y - 10 = 0$ and $5x - 2y = 0$ are perpendicular. (2 marks)

**139.** Find the length of the interval joining $B(-1, 2)$ and $C(3, 5)$. (1 mark)

**140.** Differentiate $\dfrac{1}{(2x+3)^2}$ with respect to $x$. (2 marks)

**141.** Differentiate $\dfrac{x^2}{2x+1}$ with respect to $x$. (2 marks)

**142.** Solve $\dfrac{3x+4}{x} = 5$. (2 marks)

**143.** Solve $x^2 + 2x - 2 = 0$, giving your answers in simplest surd form. (2 marks)

**144.** Find the exact value of $\theta$ such that $2\sin\theta = 1$, where $\dfrac{\pi}{2} \leq \theta \leq \pi$. (2 marks)

**145.** Find the exact values of $x$ such that $2\sin x = -1$, where $0 \leq x \leq 2\pi$. (2 marks)

**146.** State the range of $y = e^x + 2$. (1 mark)

**147.** Sketch the graph of $y = 3 - e^{-x}$, showing the $y$-intercept and the horizontal asymptote. (2 marks)

**148.** Find the inverse function of $f(x) = e^x + 1$, and state its domain. (2 marks)

**149.** Solve $|x - 2| = 7$. (2 marks)

**150.** It is estimated that 80% of students own a laptop.

(a) Two students are chosen at random. What is the probability that neither owns a laptop? (2 marks)

(b) Of the students who own a laptop, 20% have used it in a maths lesson. A student is chosen at random. What is the probability that the student owns a laptop and has used it in a maths lesson? (1 mark)

**151.** Expand and simplify $(\sqrt{5}-2)(3\sqrt{5}+1)$. (2 marks)

**152.** Simplify $\dfrac{(2x^3)^2}{4x^2}$. (1 mark)

# 4 The derivative of $\ln x$ and $\ln f(x)$

ANCHOR QUESTION

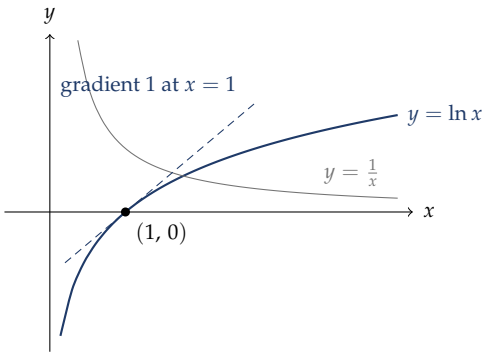
*Differentiate* $y = \ln\left(x^2 - 3x + 1\right)$.

SYLLABUS MAV-12-04

- Use chain rule on the identity $e^{\ln x} = x$ to prove the formula $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ for the derivative of the natural logarithm function where $x > 0$
- Use the chain rule to differentiate functions of the form $\ln f(x)$

NOTES

- $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$, for $x > 0$.
- Proof: differentiate both sides of $e^{\ln x} = x$. Then $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$, so $x\,\dfrac{d}{dx}(\ln x) = 1$.
- $\dfrac{d}{dx}\left(\ln f(x)\right) = \dfrac{f'(x)}{f(x)}$: the derivative of the inside, over the inside. This is on the reference sheet.
- $\ln(kx) = \ln k + \ln x$, so $\ln(kx)$ has the same derivative as $\ln x$: $\dfrac{1}{x}$.
- A product, quotient or power inside the log: expand with the log laws first. $\ln(AB) = \ln A + \ln B$, $\ln\dfrac{A}{B} = \ln A - \ln B$, $\ln A^n = n \ln A$.
- A sum or difference inside the log: differentiate it as it is.




WORKED EXAMPLE

*Differentiate* $y = \ln\left(x^2 + 4x - 1\right)$.

$f(x) = x^2 + 4x - 1$, so $f'(x) = 2x + 4$. the inside and its derivative

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{2x+4}{x^2+4x-1}$$

## Tier 1 · the log of a polynomial

**153.** For each function, write the inside $f(x)$, then find $f'(x)$.

a) $\ln(9x)$

b) $\ln\left(x^2-7\right)$

c) $\ln(5-2x)$

**154.** Expand using the log laws.

a) $\ln\left(x^3(x+2)\right)$

b) $\ln\left(\dfrac{x-1}{x+5}\right)$

c) $\ln\left((4x+3)^2\right)$

**155.** Should you use the log laws first? Answer yes or no.

a) $\ln\left((x+1)(x+2)\right)$

b) $\ln\left(2x^3+1\right)$

c) $\ln\left((x-3)^4\right)$

**156.** Differentiate:

a) $y=\ln\left(x^2+1\right)$

b) $y=\ln\left(x^3-4x\right)$

**157.** Differentiate $y=\ln\left(2x^2-5x+3\right)$.

## Tier 2 · a linear inside

**158.** Differentiate $y=\ln(6x)$.

**159.** Differentiate $y=\ln(3x+2)$.

**160.** Differentiate $y=\ln(1-4x)$.

## Tier 3 · coefficients and sums

**161.** Differentiate $y=5\ln(2x+3)$.

**162.** Differentiate $y=x^3-2\ln x$.

**163.** Differentiate $y=\ln\left(x^2+1\right)+e^{2x}$.

### Tier 4 · the power law first

**164.** Differentiate $y = \ln\left((2x+1)^3\right)$.

**165.** Differentiate $y = \ln\sqrt{4x-1}$.

**166.** Differentiate $y = \ln\left(x^5\right)$.

### Tier 5 · the product and quotient laws first

**167.** Differentiate $y = \ln\left(x(x+3)\right)$, giving your answer as a single fraction.

**168.** Differentiate $y = \ln\left(\dfrac{x+2}{x-1}\right)$, giving your answer as a single fraction.

**169.** Differentiate $y = \ln\left(\dfrac{3x-1}{x+1}\right)$, giving your answer as a single fraction.

### Tier 6 · several log laws

**170.** Differentiate $y = \ln\left(x^3\sqrt{x+1}\right)$, giving your answer as a single fraction.

**171.** Differentiate $y = \ln\left(\dfrac{(x+1)^2}{x-2}\right)$, giving your answer as a single fraction.

**172.** Differentiate $y = \ln\sqrt{\dfrac{x+1}{x-1}}$, simplifying fully.

### Conceptual

**173.** Use the identity $e^{\ln x} = x$ to prove that $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

**174.** Show that $\ln(5x)$ and $\ln x$ have the same derivative, and explain why using the log laws.

**175.** A student writes $\dfrac{d}{dx}\ln\left(x^2+1\right) = \dfrac{1}{x^2+1}$. Explain the error and give the correct derivative.

**176.** Find the exact gradient of the tangent to $y = \ln\left(x^2+3\right)$ at the point where $x = 1$.

**177.** Let $f(x) = \ln\left(x^2 - 2x + 5\right)$.

(a) Find $f'(x)$. (1 mark)

(b) Explain why $f(x)$ is defined for all real $x$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

## Revision

**178.** Differentiate $e^{3x} + e^{-x}$ with respect to $x$. (1 mark)

**179.** Find the derivative of $y = (x^2 + 1)e^{2x}$. (2 marks)

**180.** Differentiate $3 \times 5^x$ with respect to $x$. (1 mark)

**181.** Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 0$. (2 marks)

**182.** Find the equation of the normal to $y = e^x$ at the point $(0, 1)$. (2 marks)

**183.** Show that the tangent to $y = e^x$ at the point where $x = 1$ passes through the origin. (2 marks)

**184.** Evaluate $e^{2\ln 3}$. (1 mark)

**185.** Solve $\ln x = 1.5$. Give your answer correct to four decimal places. (2 marks)

**186.** Solve $\log_2(x + 1) = 3$. (1 mark)

**187.** Write $2\ln 3 + \ln 4$ as a single logarithm. (1 mark)

**188.** Solve $5e^{0.2t} = 40$, giving the exact answer. (2 marks)

**189.** Use the change of base rule to show that $\log_4 x = \frac{1}{2}\log_2 x$. (2 marks)

**190.** Differentiate $(3x^2 - 1)^5$ with respect to $x$. (2 marks)

**191.** Differentiate $x\sqrt{x}$ with respect to $x$. (1 mark)

**192.** Find the coordinates of the point on $y = x^2 - 6x$ where the tangent is parallel to the line $y = 4x + 1$. (2 marks)

**193.** Let $f(x) = x^3 + 3x^2 + kx - 1$, where $k$ is a constant. Find the values of $k$ for which $f(x)$ is an increasing function. (2 marks)

**194.** Show that the equation $x^2 + x + 3 = 0$ has no real roots. (1 mark)

**195.** The points $A$ and $B$ lie on a circle with centre $O$ and radius 2, with $\angle AOB = \theta$. The area of triangle $AOB$ is $\sqrt{3}$. One value of $\theta$ is $\frac{\pi}{3}$. Find the other value. (2 marks)

**196.** Find the exact values of $x$ such that $2\cos x = \sqrt{3}$, where $0 \leq x \leq 2\pi$. (2 marks)

**197.** What is the domain of $f(x) = \sqrt{4 - x}$? (1 mark)

**198.** Sketch the curve $y = \ln x$, showing its intercept and asymptote. (1 mark)

**199.** Solve $|2x + 3| = 9$. (2 marks)

**200.** Let $f(x) = 2x - 1$. Find $f^{-1}(x)$ and evaluate $f^{-1}(5)$. (1 mark)

**201.** The probability that Chloe wins a prize in her weekly raffle is $\frac{1}{8}$. The probability that Dev wins a prize in his is $\frac{1}{10}$. What is the probability that, in a given week, at least one of them wins a prize? (2 marks)

**202.** Simplify $\dfrac{x^2 - 4x}{x^2 - 16}$. (1 mark)

# 5 $\log_a x$, combined rules and tangents

ANCHOR QUESTION

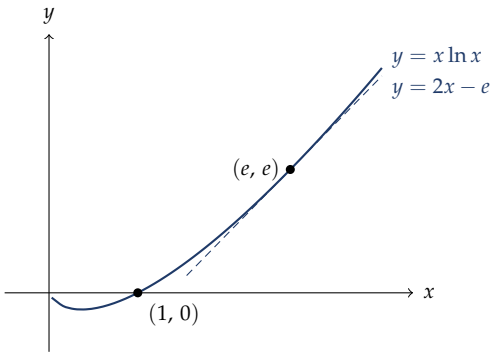
*Differentiate $y = x^2 \ln x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Prove and use the formula $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$, where $a$ is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Change of base: $\log_a x = \dfrac{\ln x}{\ln a}$.
- $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$. Proof: $\log_a x = \dfrac{1}{\ln a} \cdot \ln x$, and $\dfrac{1}{\ln a}$ is a constant.
- For $\log_a f(x)$: change to $\dfrac{\ln f(x)}{\ln a}$, then use $\dfrac{f'(x)}{f(x)}$.
- A power inside the log, as in $\ln(x^3)$: log law first, $3 \ln x$.
- A power on the whole log, as in $(\ln x)^3$: chain rule on the outside, $3(\ln x)^2 \times \dfrac{1}{x}$. Note $\ln^n x$ means $(\ln x)^n$.
- Tangents and normals at $x = e$: use $\ln e = 1$ and $\ln 1 = 0$ to keep answers exact.




WORKED EXAMPLE

*Differentiate $y = x^4 \ln x$, giving your answer in factorised form.*

$u = x^4$, $u' = 4x^3$; $\quad v = \ln x$, $v' = \dfrac{1}{x}$ — name the parts

$\dfrac{dy}{dx} = 4x^3 \ln x + x^4 \times \dfrac{1}{x} = 4x^3 \ln x + x^3$ — product rule

$= x^3(4 \ln x + 1)$ — common factor $x^3$

## Tier 1 · products with $\ln x$

**203.** Write using natural logarithms.

a) $\log_6 x$
b) $\log_3 (x^2 - 1)$

**204.** Differentiate:

a) $y = \log_4 x$
b) $y = 3\log_5 x$
c) $y = \log_2(3x - 1)$

**205.** For each function, state whether to use a log law first or the chain rule on the outside.

a) $\ln (x^6)$
b) $(\ln x)^6$
c) $\ln^2(3x)$
d) $\ln ((x+4)^3)$

**206.** Differentiate $y = x \ln x$.

**207.** Differentiate $y = x^5 \ln x$, giving your answer in factorised form.

## Tier 2 · other products with logs

**208.** Differentiate $y = e^x \ln x$.

**209.** Differentiate $y = x^2 \ln(3x)$, giving your answer in factorised form.

**210.** Differentiate $y = \sqrt{x}\, \ln x$, giving your answer as a single fraction.

## Tier 3 · the quotient rule with logs

**211.** Differentiate $y = \dfrac{\ln x}{x^2}$.

**212.** Differentiate $y = \dfrac{x}{\ln x}$.

**213.** Differentiate $y = \dfrac{\ln x}{x+1}$, giving your answer as a single fraction.

## Tier 4 · powers of $\ln x$

**214.** Differentiate $y = (\ln x)^2$.

**215.** Differentiate $y = (\ln x)^4$.

**216.** Differentiate $y = \sqrt{\ln x}$.

## Tier 5 · tangents with logs

**217.** Find the equation of the tangent to $y = \ln x$ at the point where $x = 1$.

**218.** Find the equation of the tangent to $y = \ln(3x)$ at the point where $x = \frac{1}{3}$.

**219.** Find the equation of the tangent to $y = 2\ln x$ at the point where $x = e$.

## Tier 6 · normals and combined rules

**220.** Find the equation of the normal to $y = \ln x$ at the point where $x = e$.

**221.** Find the equation of the tangent to $y = x^2 \ln x$ at the point where $x = 1$.

**222.** Find the equation of the normal to $y = x\ln x$ at the point where $x = 1$.

## Conceptual

**223.** Prove that $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x\ln a}$.

**224.** Find $\dfrac{d}{dx}\ln\left(x^2\right)$ and $\dfrac{d}{dx}(\ln x)^2$, and explain why they are different.

**225.** Show that the tangent to $y = \ln x$ at the point where $x = e$ passes through the origin.

**226.** Find the exact coordinates of the point on $y = x\ln x$ where the tangent is horizontal.

**227.** Let $f(x) = \dfrac{\ln x}{x}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**228.** Differentiate $\ln(x^2+7)$ with respect to $x$. (1 mark)

**229.** Differentiate $\ln\left(\dfrac{x+1}{x-1}\right)$, giving your answer as a single fraction. (2 marks)

**230.** Find the derivative of $y = xe^{-x}$. (2 marks)

**231.** Differentiate $10^x$ with respect to $x$. (1 mark)

**232.** Differentiate $e^{\sqrt{x}}$ with respect to $x$. (2 marks)

**233.** Find the gradient of the tangent to the curve $y = \ln(4x)$ at the point where $x = 3$. (2 marks)

**234.** Show that $\dfrac{d}{dx}\left(e^{2x} - e^{-2x}\right) = 2\left(e^{2x} + e^{-2x}\right)$. (2 marks)

**235.** Solve $e^{2x} - 3e^x + 2 = 0$. (3 marks)

**236.** Differentiate $\ln(3x-4)$ with respect to $x$. (2 marks)

**237.** Simplify $\ln(e^x) + e^{\ln x}$. (1 mark)

**238.** Evaluate $\log_4 2$. (1 mark)

**239.** Write $\ln 18 - \ln 2 + \ln 5$ as a single logarithm. (1 mark)

**240.** Solve $\log_2 x + \dfrac{3}{\log_2 x} = 4$. (3 marks)

**241.** Differentiate $\dfrac{6}{\sqrt{x}}$ with respect to $x$. (1 mark)

**242.** Find the equation of the normal to $y = x^2$ at the point $(1, 1)$. Give your answer in general form. (2 marks)

**243.** Solve $x^3 - 4x = 0$. (1 mark)

**244.** Find the values of $k$ for which $x^2 + kx + 9 = 0$ has equal roots. (2 marks)

**245.** Find the exact values of $x$ such that $2\sin x = -\sqrt{3}$, where $0 \leq x \leq 2\pi$. (2 marks)

**246.** Solve $2\sin^2 x = 1$ for $0 \leq x \leq 2\pi$. (3 marks)

**247.** The graph of $y = A\sin bx$ has amplitude 3 and period $\pi$. Find the values of $A$ and $b$, where $A > 0$ and $b > 0$. (2 marks)

**248.** Shade the region in the plane defined by $y \geq 0$ and $y \leq 9 - x^2$. (2 marks)

**249.** Solve $|x+1| \geq 3$. (2 marks)

**250.** Given $f(x) = \sqrt{x}$ and $g(x) = x^2 + 9$, find $f(g(4))$. (1 mark)

**251.** There are twelve chocolates in a box: four mint, four caramel and four strawberry. Two are chosen at random and eaten.

(a) What is the probability that the two chocolates have the same centre? (1 mark)

(b) What is the probability that they have different centres? (1 mark)

**252.** Factorise $x^3 - 8$. (1 mark)

# 6 Derivatives of $\sin x$, $\cos x$ and $\tan x$

ANCHOR QUESTION

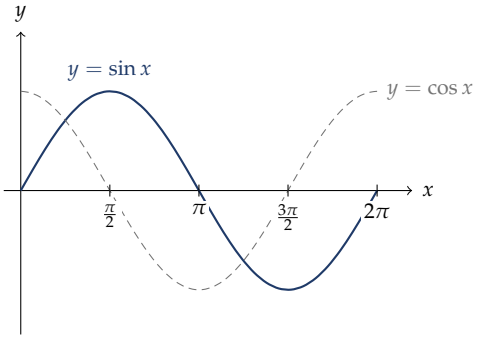
*Differentiate* $y = \sin\left(x^2 + 2x\right)$.

SYLLABUS MAV-12-04

- Determine the formulas $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ informally by using the graphs of $y = \sin x$ and $y = \cos x$ and verify with graphing applications
- Use the rules of differentiation to show that $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the chain rule to differentiate functions of the form $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$

NOTES

- $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$. Angles must be in radians.
- From the graph of $y = \sin x$: the gradient is 0 at the turning points $x = \frac{\pi}{2}$, $\frac{3\pi}{2}$, is 1 at $x = 0$ and $-1$ at $x = \pi$. That is the graph of $y = \cos x$.
- $\frac{d}{dx}(\tan x) = \sec^2 x$, where $\sec x = \frac{1}{\cos x}$. Proof: write $\tan x = \frac{\sin x}{\cos x}$ and use the quotient rule with $\sin^2 x + \cos^2 x = 1$.
- Chain rule: $\frac{d}{dx}\sin f(x) = f'(x)\cos f(x)$, $\frac{d}{dx}\cos f(x) = -f'(x)\sin f(x)$, $\frac{d}{dx}\tan f(x) = f'(x)\sec^2 f(x)$.
- The angle stays the same; multiply by the derivative of the angle.




WORKED EXAMPLE

*Differentiate* $y = \cos\left(x^2 + 4x\right)$.

$f(x) = x^2 + 4x$, so $f'(x) = 2x + 4$. — the angle and its derivative

$\frac{dy}{dx} = -f'(x)\sin f(x)$ — cos gives − sin

$= -(2x + 4)\sin\left(x^2 + 4x\right)$

## Tier 1 · $\sin f(x)$

**253.** For each function, write the angle $f(x)$, then find $f'(x)$.

a) $\sin(9x)$

b) $\cos\left(x^2 - 1\right)$

c) $\tan(5 - 2x)$

**254.** Differentiate:

a) $y = \sin x + \cos x$

b) $y = 4\tan x$

c) $y = x^2 - 3\cos x$

**255.** Differentiate $y = \sin\left(x^2 + 3x\right)$.

**256.** Differentiate $y = \sin\left(x^3\right)$.

**257.** Differentiate $y = \sin(5x - 2)$.

## Tier 2 · $\cos f(x)$

**258.** Differentiate $y = \cos\left(x^2 + 1\right)$.

**259.** Differentiate $y = \cos 4x$.

**260.** Differentiate $y = \cos(2 - 3x)$.

## Tier 3 · $\tan f(x)$

**261.** Differentiate $y = \tan 6x$.

**262.** Differentiate $y = \tan\left(x^2\right)$.

**263.** Differentiate $y = \tan(1 - x)$.

## Tier 4 · coefficients and sums

**264.** Differentiate $y = 5\sin 3x + 2\cos 4x$.

**265.** Differentiate $y = 2\tan\dfrac{x}{2}$.

**266.** Differentiate $y = \cos(\pi x) - \sin(\pi x)$.

## Tier 5 · exact gradients

**267.** Find the exact gradient of the tangent to $y = \cos 2x$ at the point where $x = \dfrac{\pi}{12}$.

**268.** Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{4}$.

**269.** Find the exact gradient of the tangent to $y = \sin 3x$ at the point where $x = \dfrac{\pi}{9}$.

## Tier 6 · horizontal tangents

**270.** Find the values of $x$, for $0 \leq x \leq 2\pi$, where the tangent to $y = \sin x - \cos x$ is horizontal.

**271.** Find the values of $x$, for $0 \leq x \leq 2\pi$, where the tangent to $y = x + 2\cos x$ is horizontal.

**272.** Find the values of $x$, for $0 \leq x \leq \pi$, where the tangent to $y = \sin 2x$ is horizontal.

## Conceptual

**273.** Use the quotient rule to show that $\dfrac{d}{dx}(\tan x) = \sec^2 x$.

**274.** Use the graph of $y = \sin x$ to explain why its gradient is 1 at $x = 0$, 0 at $x = \dfrac{\pi}{2}$ and $-1$ at $x = \pi$, and why this matches $y = \cos x$.

**275.** A student writes $\dfrac{d}{dx}(\cos 3x) = -\sin 3x$. Explain the error and give the correct derivative.

**276.** Explain why the graph of $y = \tan x$ has no horizontal tangents.

**277.** Let $f(x) = \sin x + \sqrt{3}\cos x$, for $0 \leq x \leq 2\pi$.

(a) Find $f'(x)$. (1 mark)

(b) Find the values of $x$ where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**278.** Differentiate $\ln(5x - 2)$ with respect to $x$. (2 marks)

**279.** Differentiate $x^3 \log_e x$ with respect to $x$. (2 marks)

**280.** Differentiate $\log_3 x$ with respect to $x$. (1 mark)

**281.** Differentiate $(\ln x)^2$ with respect to $x$. (2 marks)

**282.** Differentiate $e^{-3x}$ with respect to $x$. (1 mark)

**283.** Show that $\dfrac{d}{dx}\left[(2 + x)\ln(2 + x) - x\right] = \ln(2 + x)$. (2 marks)

**284.** Find the equation of the tangent to $y = \ln x$ at the point where $x = e^2$. (2 marks)

**285.** Find the exact value of $\sin \dfrac{5\pi}{4} + \cos \dfrac{7\pi}{4}$. (1 mark)

**286.** An arc of a circle with radius 4 cm has length 7 cm. Find the angle the arc subtends at the centre, in radians. (1 mark)

**287.** Find the exact value of $\tan \dfrac{2\pi}{3}$. (1 mark)

**288.** Find the exact values of $x$ such that $2\cos x = -1$, where $0 \leq x \leq 2\pi$. (2 marks)

**289.** Solve $\tan x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. (2 marks)

**290.** Consider the expression $\sec^2 x + \sec x \tan x$.

(a) Prove that $\sec^2 x + \sec x \tan x = \dfrac{1 + \sin x}{\cos^2 x}$. (1 mark)

(b) Hence prove that $\sec^2 x + \sec x \tan x = \dfrac{1}{1 - \sin x}$. (1 mark)

**291.** State the amplitude and the period of $y = 3\cos 2x$. (2 marks)

**292.** Find the exact length of an arc of a circle with radius 6 cm that subtends an angle of $\dfrac{\pi}{3}$ at the centre. (1 mark)

**293.** The points $A$ and $B$ lie on a circle with centre $O$ and radius 3 cm, and $\angle AOB = \dfrac{\pi}{3}$.

(a) Find the exact area of the sector $AOB$. (1 mark)

(b) Find the exact perimeter of the minor segment bounded by the chord $AB$ and the arc $AB$. (2 marks)

**294.** Solve $4^{x-1} = 32$. (2 marks)

**295.** Solve $\ln x + \ln 2 = \ln 10$. (1 mark)

**296.** Differentiate $(x^2 - 3)^4$ with respect to $x$. (2 marks)

**297.** Find the equation of the normal to $y = x^3$ at the point $(1, 1)$. Give your answer in general form. (2 marks)

**298.** Solve the inequality $x^2 + 4x - 5 < 0$. (2 marks)

**299.** Show that $f(x) = x \sin x$ is an even function. (1 mark)

**300.** Xena and Gabrielle play a series of games. The series finishes when one player has won two games. In any game, the probability that Xena wins is $\dfrac{3}{4}$ and the probability that Gabrielle wins is $\dfrac{1}{4}$.

(a) What is the probability that Gabrielle wins the series? (2 marks)

(b) What is the probability that three games are played? (2 marks)

**301.** Rationalise the denominator of $\dfrac{6}{\sqrt{7} - \sqrt{5}}$. Give your answer in simplest form. (2 marks)

**302.** Write down the equation of the circle with centre $(3, -2)$ and radius 4. (1 mark)

# 7 Trig functions with combined rules

ANCHOR QUESTION

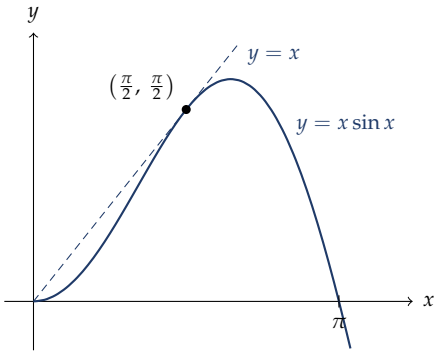
*Differentiate $y = x^2 \sin x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sin^n x$ means $(\sin x)^n$: use the chain rule on the outside, e.g. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$.
- Product rule $u'v + uv'$ and quotient rule $\dfrac{u'v - uv'}{v^2}$ work with trig functions exactly as before. Name $u$, $v$, $u'$, $v'$ first.
- Take out common factors, e.g. $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$.
- Trig inside $e$ or ln: $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$ and $\dfrac{d}{dx}\ln(\cos x) = \dfrac{-\sin x}{\cos x} = -\tan x$.
- For tangents, use exact values: $\sin\dfrac{\pi}{2} = 1$, $\cos\dfrac{\pi}{2} = 0$, $\cos\pi = -1$.




WORKED EXAMPLE

*Differentiate $y = x^3\cos x$, giving your answer in factorised form.*

$u = x^3$, $u' = 3x^2$; $\quad v = \cos x$, $v' = -\sin x$ — name the parts

$\dfrac{dy}{dx} = 3x^2\cos x - x^3\sin x$ — product rule

$= x^2(3\cos x - x\sin x)$ — common factor $x^2$

## Tier 1 · products with $\sin x$, $\cos x$, $\tan x$

**303.** Rewrite with the power outside a bracket.

a) $\cos^3 x$

b) $\sin^2 4x$

**304.** Differentiate $y = x \sin x$.

**305.** Differentiate $y = x^2 \cos x$, giving your answer in factorised form.

**306.** Differentiate $y = x^3 \tan x$, giving your answer in factorised form.

**307.** Differentiate $y = \sqrt{x}\, \sin x$.

## Tier 2 · products with a chain inside

**308.** Differentiate $y = x \sin 3x$.

**309.** Differentiate $y = x^2 \cos 2x$, giving your answer in factorised form.

**310.** Differentiate $y = \sin x \cos x$.

## Tier 3 · powers of trig functions

**311.** Differentiate $y = \cos^2 x$.

**312.** Differentiate $y = \sin^3 x$.

**313.** Differentiate $y = \tan^2 x$.

## Tier 4 · the quotient rule with trig

**314.** Differentiate $y = \dfrac{\cos x}{x}$.

**315.** Differentiate $y = \dfrac{x}{\sin x}$.

**316.** Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{1+\cos x}\right) = \dfrac{1}{1+\cos x}$.

## Tier 5 · trig with $e$ and $\ln$

**317.** Differentiate $y = e^x \sin x$, giving your answer in factorised form.

**318.** Differentiate $y = \ln(\sin x)$.

**319.** Differentiate $y = e^{\cos x}$.

## Tier 6 · tangents and normals

**320.** Find the equation of the tangent to $y = x\cos x$ at the point where $x = \pi$.

**321.** Find the equation of the tangent to $y = \sin^2 x$ at the point where $x = \dfrac{\pi}{4}$.

**322.** Find the equation of the normal to $y = e^x \cos x$ at the point where $x = 0$.

## Tier 7 · show that

**323.** Show that $\dfrac{d}{dx}\left(\dfrac{\cos x}{1-\sin x}\right) = \dfrac{1}{1-\sin x}$.

**324.** Show that $\dfrac{d}{dx}(x\sin x + \cos x) = x\cos x$.

**325.** Show that $\dfrac{d}{dx}\left[e^{-x}(\sin x + \cos x)\right] = -2e^{-x}\sin x$.

## Conceptual

**326.** Show that $\dfrac{d}{dx}\left(\sin^2 x + \cos^2 x\right) = 0$, and explain why this makes sense.

**327.** Show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$.

**328.** A student writes $\dfrac{d}{dx}(x\sin x) = \cos x$. Explain the error and give the correct derivative.

**329.** Find the exact gradient of the tangent to $y = x^2\sin x$ at the point where $x = \pi$.

**330.** Let $f(x) = e^{-x}\sin x$, for $0 \le x \le 2\pi$.

(a) Show that $f'(x) = e^{-x}(\cos x - \sin x)$. (1 mark)

(b) Find the values of $x$ where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**331.** Differentiate $\sin(x^2+1)$ with respect to $x$. (2 marks)

**332.** Differentiate $\tan 5x$ with respect to $x$. (1 mark)

**333.** Differentiate $\cos(3x+1)$ with respect to $x$. (1 mark)

**334.** Find the exact gradient of the tangent to $y = \cos x$ at the point where $x = \dfrac{\pi}{6}$. (2 marks)

**335.** Differentiate $e^{x^2}$ with respect to $x$. (1 mark)

**336.** Differentiate $\ln(x^3+1)$ with respect to $x$. (1 mark)

**337.** Differentiate $2^{3x}$ with respect to $x$. (1 mark)

**338.** Differentiate $\log_{10} x$ with respect to $x$. (1 mark)

**339.** Find the derivative of $y = x^2 e^x$. (2 marks)

**340.** Differentiate $\dfrac{\ln x}{x}$ with respect to $x$. (2 marks)

**341.** Find the equation of the tangent to $y = e^x$ at the point where $x = 1$. (2 marks)

**342.** Find the equation of the normal to $y = \ln x$ at the point where $x = 1$. (2 marks)

**343.** Solve $2\cos^2 \dfrac{x}{3} = 1$ for $-\pi \le x \le \pi$. (3 marks)

**344.** Solve $\sin 2x = 0$ for $0 \le x \le \pi$. (2 marks)

**345.** Find the exact values of $x$ such that $2\cos x + \sqrt{3} = 0$, where $0 \le x \le 2\pi$. (2 marks)

**346.** Prove that $1 + \tan^2 x = \sec^2 x$. (2 marks)

**347.** State the range of $y = 2\sin x + 1$. (1 mark)

**348.** State the period of $y = \tan 2x$. (1 mark)

**349.** Sketch $y = 3\sin x + 1$ for $0 \le x \le \pi$. (2 marks)

**350.** Simplify $\ln(e^2 x) - \ln x$. (1 mark)

**351.** Evaluate $\log_2 48 - \log_2 3$. (1 mark)

**352.** Differentiate $(1-2x)^5$ with respect to $x$. (2 marks)

**353.** Differentiate $\dfrac{x^2}{x+1}$ with respect to $x$. (2 marks)

**354.** Solve $x^2 - 2x - 1 = 0$, giving your answers in simplest surd form. (2 marks)

**355.** Find the gradient of a line perpendicular to $3x + 4y = 12$. (1 mark)

**356.** Let $f(x) = x^2 - 1$. Find and simplify $f(a+1)$. (1 mark)

**357.** Ana has two red shirts and three yellow shirts. On each of Monday, Tuesday and Wednesday she chooses one shirt at random to wear, and she wears each shirt only once. What is the probability that she does not wear the same colour on consecutive days? (2 marks)

**358.** Simplify $\sqrt{50} - \sqrt{18}$. (1 mark)

# 8 Derivatives of the reciprocal trig functions

ANCHOR QUESTION

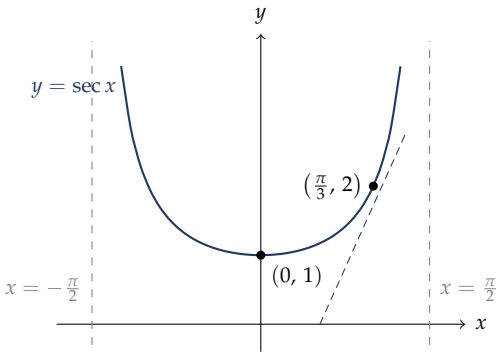
*Show that* $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

SYLLABUS MAV-12-04

- Use the rules of differentiation to find the derivatives of cosec $x$, sec $x$ and cot $x$
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sec x = \dfrac{1}{\cos x} = (\cos x)^{-1}$, $\operatorname{cosec} x = \dfrac{1}{\sin x} = (\sin x)^{-1}$, $\cot x = \dfrac{\cos x}{\sin x}$.
- $\dfrac{d}{dx}(\sec x) = \tan x \sec x$: chain rule on $(\cos x)^{-1}$.
- $\dfrac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$: chain rule on $(\sin x)^{-1}$.
- $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$: quotient rule on $\dfrac{\cos x}{\sin x}$.
- These three are *not* on the reference sheet. The "co" functions get the minus sign.
- With an angle $f(x)$, multiply by $f'(x)$ as usual, e.g. $\dfrac{d}{dx}\sec 2x = 2\tan 2x \sec 2x$.




WORKED EXAMPLE

*Show that* $\dfrac{d}{dx}(\sec 4x) = 4\tan 4x \sec 4x$.

$y = (\cos 4x)^{-1}$ — write as a power

$\dfrac{dy}{dx} = -(\cos 4x)^{-2} \times (-4\sin 4x)$ — chain rule

$= \dfrac{4\sin 4x}{\cos^2 4x} = 4 \cdot \dfrac{\sin 4x}{\cos 4x} \cdot \dfrac{1}{\cos 4x} = 4\tan 4x \sec 4x$

## Tier 1 · $\sec x$ as a power

**359.** Write as a power of a trig function.

a) $\operatorname{cosec} 5x$

b) $\sec\left(x^3\right)$

**360.** Find the exact value.

a) $\sec \dfrac{\pi}{6}$

b) $\operatorname{cosec} \dfrac{\pi}{3}$

c) $\cot \dfrac{\pi}{4}$

d) $\operatorname{cosec} \dfrac{\pi}{2}$

**361.** By writing $\sec 3x = (\cos 3x)^{-1}$, show that $\dfrac{d}{dx}(\sec 3x) = 3\tan 3x \sec 3x$.

**362.** Differentiate $y = \sec^2 x$.

**363.** Differentiate $y = 5\sec x$.

## Tier 2 · $\operatorname{cosec} x$

**364.** Differentiate $y = \operatorname{cosec} x + x^2$.

**365.** Differentiate $y = 2\operatorname{cosec} x$.

**366.** Show that $\dfrac{d}{dx}(\operatorname{cosec} 2x) = -2\cot 2x \operatorname{cosec} 2x$.

## Tier 3 · $\cot x$

**367.** Differentiate $y = \cot x - x$.

**368.** Differentiate $y = 4\cot x$.

**369.** By writing $\cot x = (\tan x)^{-1}$, show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

## Tier 4 · sums

**370.** Differentiate $y = \sec x + \operatorname{cosec} x$.

**371.** Differentiate $y = 2\cot x - 3\sec x$.

**372.** Differentiate $y = \tan x + \cot x$.

## Tier 5 · the chain rule

**373.** Differentiate $y = \sec\left(x^2\right)$.

**374.** Differentiate $y = \cot 3x$.

**375.** Differentiate $y = \operatorname{cosec}(2x - 1)$.

## Tier 6 · the product rule

**376.** Differentiate $y = x \operatorname{cosec} x$, giving your answer in factorised form.

**377.** Differentiate $y = x^2 \cot x$, giving your answer in factorised form.

**378.** Differentiate $y = e^x \sec x$, giving your answer in factorised form.

## Tier 7 · tangents and normals

**379.** Find the equation of the tangent to $y = \sec x$ at the point where $x = \dfrac{\pi}{3}$.

**380.** Find the gradient of the normal to $y = \operatorname{cosec} x$ at the point where $x = \dfrac{\pi}{4}$.

**381.** Find the equation of the tangent to $y = \cot x$ at the point where $x = \dfrac{\pi}{2}$.

## Conceptual

**382.** Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

**383.** Use the quotient rule to show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

**384.** Explain why $y = \cot x$ is decreasing wherever it is defined.

**385.** A student says that since $\sec x = \dfrac{1}{\cos x}$, its derivative is $\dfrac{1}{-\sin x}$. Explain the error.

**386.** Let $f(x) = \sec x$, for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

(a) Show that $f'(x) = \dfrac{\sin x}{\cos^2 x}$. (1 mark)

(b) Find the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

(c) Explain why $f'(x) > 0$ for $0 < x < \dfrac{\pi}{2}$. (1 mark)

## Revision

**387.** Differentiate $x^2 \sin x$ with respect to $x$. (2 marks)

**388.** Differentiate $\cos^2 x$ with respect to $x$. (2 marks)

**389.** Differentiate $x \tan 2x$ with respect to $x$. (2 marks)

**390.** Differentiate $e^{\sin x}$ with respect to $x$. (1 mark)

**391.** Differentiate $\dfrac{\sin 2x}{x}$ with respect to $x$. (2 marks)

**392.** Differentiate $\log_e(\sin x)$ with respect to $x$. (2 marks)

**393.** Differentiate $\dfrac{\cos x}{x+2}$ with respect to $x$. (2 marks)

**394.** Find the derivative of $y = x^2 e^{-x}$. (2 marks)

**395.** Find the equation of the tangent to $y = \sin x$ at the point where $x = \pi$. (2 marks)

**396.** Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{3}$. (2 marks)

**397.** Solve $\sin x = \cos x$ for $0 \leq x \leq 2\pi$. (2 marks)

**398.** Differentiate $\dfrac{x^2}{\cos x}$ with respect to $x$. (2 marks)

**399.** Find the exact value of $\sin \dfrac{4\pi}{3}$. (1 mark)

**400.** Show that $(1 - \cos^2 x)\operatorname{cosec} x = \sin x$. (1 mark)

**401.** Simplify $1 + \cot^2 x$. (1 mark)

**402.** Find the exact values of $x$ such that $2\sin x - 1 = 0$, where $0 \leq x \leq 2\pi$. (2 marks)

**403.** Solve $\ln(x-2) = \ln 5$. (1 mark)

**404.** Solve $3e^{2x} = 12$, giving the exact answer. (2 marks)

**405.** Evaluate $\log_3 5 + \log_3 \frac{1}{5}$. (1 mark)

**406.** Differentiate $3x^4 - \dfrac{2}{x}$ with respect to $x$. (1 mark)

**407.** Differentiate $\sqrt{x^2+4}$ with respect to $x$. (2 marks)

**408.** Show that the curve $y = x^3 + 3x$ has no horizontal tangents. (2 marks)

**409.** Solve $x^2 > 9$. (2 marks)

**410.** The line $y = 2x + c$ is a tangent to the parabola $y = x^2 + 3$. Find the value of $c$. (2 marks)

**411.** Solve simultaneously: $y = x^2$ and $y = x + 2$. (2 marks)

**412.** What is the domain of $y = \ln(x^2)$? (1 mark)

**413.** James parks his car on one of four levels of a car park, chosen at random. When he leaves, he chooses a level at random and searches it. If his car is not there, he chooses a different level, and continues until he finds it.

(a) What is the probability that his car is on the first level he searches? (1 mark)

(b) What is the probability that he must search all four levels? (1 mark)

(c) What is the probability that, on each of the five working days in a week, his car is not on the first level he searches? (1 mark)

**414.** Simplify $(3a^2b)^3$. (1 mark)

# 9 Mixed differentiation: choosing the rule

ANCHOR QUESTION

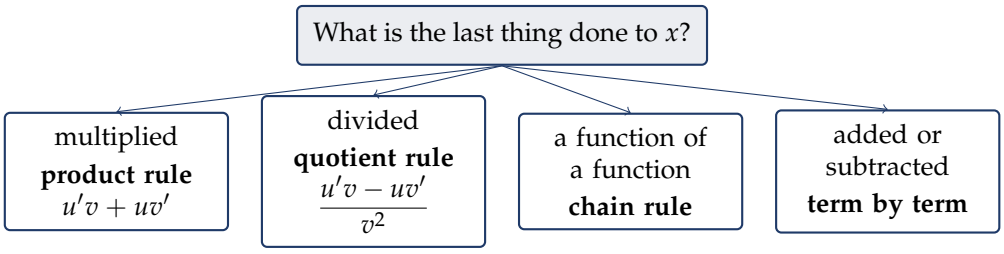
*Differentiate $y = e^{2x}\sin 3x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Standard derivatives: $e^x \to e^x$, $\ln x \to \dfrac{1}{x}$, $a^x \to (\ln a)\,a^x$, $\sin x \to \cos x$, $\cos x \to -\sin x$, $\tan x \to \sec^2 x$, $\sec x \to \tan x \sec x$, $\operatorname{cosec} x \to -\cot x \operatorname{cosec} x$, $\cot x \to -\operatorname{cosec}^2 x$.
- Choose the rule from the *last* thing done to $x$: multiplied, divided, a function of a function, or added.
- Inside a product or quotient, each part may need its own chain rule, e.g. $\dfrac{d}{dx}\sin 3x = 3\cos 3x$.
- Simplify by taking out common factors, especially exponentials.
- $e^{f(x)}$, $\ln f(x)$, $\sin f(x)$ and $[f(x)]^n$ all follow the chain rule: keep the inside, multiply by its derivative.




WORKED EXAMPLE

*Differentiate $y = e^{-x}\sin 2x$, giving your answer in factorised form.*

Last operation: multiplied, so the product rule. — choose the rule

$u = e^{-x}$, $u' = -e^{-x}$; $\quad v = \sin 2x$, $v' = 2\cos 2x$ — chain rule on each part

$$\frac{dy}{dx} = -e^{-x}\sin 2x + 2e^{-x}\cos 2x$$

$$= e^{-x}(2\cos 2x - \sin 2x)$$ — common factor $e^{-x}$

## Tier 1 · products with a chain inside

**415.** Differentiate:

a) $e^x$

b) $\ln x$

c) $\sin x$

d) $\cot x$

e) $3^x$

**416.** Which rule comes first?

a) $x^2 \cos x$

b) $\cos\left(x^2\right)$

c) $\dfrac{\cos x}{x^2}$

d) $x^2 + \cos x$

**417.** Differentiate $y = e^x \cos 2x$, giving your answer in factorised form.

**418.** Differentiate $y = x^2 e^{3x}$, giving your answer in factorised form.

## Tier 2 · products with a negative power

**419.** Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

**420.** Differentiate $y = xe^{-3x}$, giving your answer in factorised form.

**421.** Differentiate $y = e^{-x} \cos x$, giving your answer in factorised form.

## Tier 3 · quotients

**422.** Differentiate $y = \dfrac{\ln x}{e^x}$, giving your answer as a single fraction.

**423.** Differentiate $y = \dfrac{\sin x}{e^{2x}}$.

**424.** Differentiate $y = \dfrac{x^2}{\ln x}$, giving your answer in factorised form.

## Tier 4 · a function of a function

**425.** Differentiate $y = \cos\left(e^x\right)$.

**426.** Differentiate $y = e^{\tan x}$.

**427.** Differentiate $y = \ln\left(1 + e^x\right)$.

### Tier 5 · the log of a sum

**428.** Differentiate $y = \ln(x + \cos x)$.

**429.** Differentiate $y = \ln\left(e^{x} + e^{-x}\right)$.

**430.** Differentiate $y = \ln(2x + \sin 2x)$.

### Tier 6 · powers of functions

**431.** Differentiate $y = (e^{x} + 1)^{3}$.

**432.** Differentiate $y = (\sin x + \cos x)^{2}$.

**433.** Differentiate $y = \sqrt{1 + \ln x}$.

### Tier 7 · quotients with a chain

**434.** Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{(1+\cos x)^{2}}\right) = \dfrac{2 - \cos x}{(1+\cos x)^{2}}$.

**435.** Differentiate $y = \dfrac{x}{\left(x^{2}+1\right)^{2}}$, simplifying fully.

**436.** Differentiate $y = \dfrac{e^{x}}{\left(e^{x}+1\right)^{3}}$, simplifying fully.

### Tier 8 · tangents and normals

**437.** Find the equation of the tangent to $y = e^{x}\sin x$ at the origin.

**438.** Find the equation of the tangent to $y = \ln\left(1 + x^{2}\right)$ at the point where $x = 1$.

**439.** Find the equation of the normal to $y = xe^{2x}$ at the origin.

### Conceptual

**440.** Explain why $y = e^{x}\sin x$ needs the product rule but $y = e^{\sin x}$ needs the chain rule, and differentiate both.

**441.** Show that $\dfrac{d}{dx}\ln(\sec x) = \tan x$.

**442.** A student writes $\dfrac{d}{dx}e^{x^{2}} = x^{2}e^{x^{2}-1}$. Explain the error and give the correct derivative.

**443.** Find the exact gradient of the tangent to $y = x\ln x - x$ at the point where $x = e$.

**444.** Let $f(x) = \dfrac{\ln x}{x^2}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - 2\ln x}{x^3}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**445.** Differentiate $e^{5x-1}$ with respect to $x$. (1 mark)

**446.** Differentiate $\ln(3x^2)$ with respect to $x$. (1 mark)

**447.** Differentiate $\log_2(x+1)$ with respect to $x$. (1 mark)

**448.** Differentiate $7^x$ with respect to $x$. (1 mark)

**449.** Differentiate $\tan(x^2)$ with respect to $x$. (1 mark)

**450.** Differentiate $\sec 3x$ with respect to $x$. (1 mark)

**451.** Differentiate $\operatorname{cosec} x$ with respect to $x$. (1 mark)

**452.** Differentiate $\cot 2x$ with respect to $x$. (1 mark)

**453.** Differentiate $\sin^2 3x$ with respect to $x$. (2 marks)

**454.** Differentiate $x\cos x$ with respect to $x$. (2 marks)

**455.** Find the equation of the tangent to $y = e^{-x}$ at the point where $x = 0$. (2 marks)

**456.** Find the equation of the normal to $y = \sin x$ at the origin. (2 marks)

**457.** Show that $\dfrac{d}{dx}\left(\dfrac{e^x - 1}{e^x + 1}\right) = \dfrac{2e^x}{(e^x+1)^2}$. (2 marks)

**458.** Let $g(x) = \ln(1 + x)$.

(a) Find $g'(x)$. (1 mark)

(b) Hence show that $g'(x) \leq 1$ for all $x \geq 0$. (1 mark)

**459.** Find the exact value of $\sec \dfrac{\pi}{4}$. (1 mark)

**460.** Solve $\tan x = -1$ for $0 \leq x \leq 2\pi$. (2 marks)

**461.** Solve $\cos 2x = 0$ for $0 \leq x \leq \pi$. (2 marks)

**462.** Solve $e^x = 3$, giving the exact answer. (1 mark)

**463.** Solve $\log_5 x = 2$. (1 mark)

**464.** Simplify $\ln\left(\dfrac{e^3}{e}\right)$. (1 mark)

**465.** Differentiate $(3x+2)^{-2}$ with respect to $x$. (1 mark)

**466.** Differentiate $\dfrac{x^2}{1-x}$ with respect to $x$. (2 marks)

**467.** Find the equation of the tangent to $y = x^2 + 3x$ at the point where $x = -1$. (2 marks)

**468.** Solve $2x^2 - 7x + 3 = 0$. (1 mark)

**469.** Let $f(x) = x^4 - 2x^2$.

(a) Find the coordinates of the points where the graph of $y = f(x)$ crosses the axes. (2 marks)

(b) Show that $f(x)$ is an even function. (1 mark)

**470.** Solve $|3x - 1| = 5$. (2 marks)

**471.** What is the domain of $f(x) = \sqrt{2x - 6}$? (1 mark)

**472.** Two identical biased coins are tossed together. The probability that both land showing heads is 0.49. What is the probability that both land showing tails? (2 marks)

**473.** The points $A(-1, 1)$ and $B(2, 4)$ lie on the parabola $y = x^2$.

(a) Find the equation of the tangent to the parabola at $A$. (2 marks)

(b) Find the coordinates of the point $C$ on the parabola where the tangent is parallel to $AB$. (2 marks)

**474.** Simplify $(x^{-2})^3 \times x^7$. (1 mark)

## Solutions

**1.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = 2 - x$, $f'(x) = -1$ c) $f(x) = x^4 - 3x$, $f'(x) = 4x^3 - 3$

**2.** a) $7e^{7x}$ b) $2x\,e^{x^2+1}$ c) $(3x^2 - 6)\,e^{x^3-6x} = 3(x^2 - 2)\,e^{x^3-6x}$

**3.** a) $-5e^{1-5x}$ b) $\dfrac{1}{2\sqrt{x}}\,e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$ c) 0, since $e^4$ is a constant

**4.** a) $12e^{3x}$ b) $-10e^{5x+1}$ c) $3e^{6x}$

**5.** a) $6x\,e^{x^2}$ b) $-(4x - 1)\,e^{2x^2-x} = (1 - 4x)\,e^{2x^2-x}$

**6.** $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = \frac{1}{4} \times 8e^{8x-2} = 2e^{8x-2}$

**7.** a) $4e^{4x} - 4e^{-4x}$ b) $6e^{2x} - 3x^2$

**8.** $\dfrac{dy}{dx} = -6e^{-3x} - 5e^x + 7$

**9.** $\dfrac{dy}{dx} = 2x - 4 - e^{2-x}$

**10.** a) $y = e^{-x}$, so $\dfrac{dy}{dx} = -e^{-x}$ b) $y = 4e^{-2x}$, so $\dfrac{dy}{dx} = -8e^{-2x}$

**11.** a) $y = e^{4x}$, so $\dfrac{dy}{dx} = 4e^{4x}$ b) $y = e^{6x}$, so $\dfrac{dy}{dx} = 6e^{6x}$

**12.** a) $y = e^{2x}$, so $\dfrac{dy}{dx} = 2e^{2x}$ b) $y = e^{2x} + 2 + e^{-2x}$, so $\dfrac{dy}{dx} = 2e^{2x} - 2e^{-2x}$

**13.** $f'(x) = 2e^{2x+1}$, so $f'(0) = 2e$.

**14.** $f'(x) = 2x\,e^{x^2-4}$, so $f'(2) = 4e^0 = 4$.

**15.** $\dfrac{dy}{dx} = -3e^{-x} + e^x$. At $x = \ln 3$: $e^{\ln 3} = 3$ and $e^{-\ln 3} = \frac{1}{3}$, so the gradient is $-3 \times \frac{1}{3} + 3 = 2$.

**16.** $\dfrac{dy}{dx} = 3\left(e^x + 1\right)^2 \times e^x = 3e^x\left(e^x + 1\right)^2$

**17.** $\dfrac{dy}{dx} = 4\left(2 - e^{3x}\right)^3 \times \left(-3e^{3x}\right) = -12e^{3x}\left(2 - e^{3x}\right)^3$

**18.** $y = \left(1 + e^{2x}\right)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(1 + e^{2x}\right)^{-\frac{1}{2}} \times 2e^{2x} = \dfrac{e^{2x}}{\sqrt{1 + e^{2x}}}$

**19.** a) $y = e^{x^2-2x}$, so $\dfrac{dy}{dx} = (2x - 2)\,e^{x^2-2x} = 2(x - 1)\,e^{x^2-2x}$. b) $e^{x^2-2x} > 0$, so $x = 1$; $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**20.** a) $y = e^{2x^2-8x}$, so $\dfrac{dy}{dx} = (4x - 8)\,e^{2x^2-8x} = 4(x - 2)\,e^{2x^2-8x}$. b) $e^{2x^2-8x} > 0$, so $x = 2$; $y = e^{8-16} = e^{-8}$. The point is $\left(2,\, e^{-8}\right)$.

**21.** a) $f'(x) = (3x^2 - 3)\,e^{x^3-3x} = 3(x-1)(x+1)\,e^{x^3-3x}$ b) $x = 1$ or $x = -1$. $f(1) = e^{-2}$ and $f(-1) = e^2$, giving $(1,\, e^{-2})$ and $(-1,\, e^2)$.

**22.** The student has used the power rule, which only applies when $x$ is in the base. Here $x$ is in the power, so the power stays the same and is multiplied by its derivative: $\dfrac{d}{dx}\left(e^{3x}\right) = 3e^{3x}$.

**23.** Let $u = ax + b$, so $y = e^u$. Then $\dfrac{du}{dx} = a$ and $\dfrac{dy}{du} = e^u$, so $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx} = e^u \times a = ae^{ax+b}$.

**24.** $\dfrac{dy}{dx} = 10e^{2x}$. Since $e^{2x} > 0$ for all $x$, the gradient is always positive, so the graph is always increasing.

**25.** $\dfrac{dy}{dx} = (2x + k)\,e^{x^2+kx}$. The exponential is never 0, so $2(3) + k = 0$ and $k = -6$.

**26.** a) $\dfrac{dy}{dx} = 10e^{2x}$ b) $10e^{2x} = 2 \times 5e^{2x} = 2y$ c) $\dfrac{dy}{dx} = 2 \times 10 = 20$

**27.** $\dfrac{d}{dx}(x^2+5)^7 = 7(x^2+5)^6 \times 2x = 14x(x^2+5)^6$ *cf. 2008 HSC Q2(a)(i)*

**28.** $\dfrac{dy}{dx} = 4x^3 - 2$. At $x = 1$: $4 - 2 = 2$. *cf. 2009 HSC Q1(d)*

**29.** Product rule: $f'(x) = (2x-1)^4 + x \times 4(2x-1)^3 \times 2 = (2x-1)^3\,[(2x-1) + 8x] = (2x-1)^3(10x - 1)$

**30.** Quotient rule: $\dfrac{3(x^2+1) - 3x \times 2x}{(x^2+1)^2} = \dfrac{3 - 3x^2}{(x^2+1)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**31.** Point $(1,\, 1)$. $\dfrac{dy}{dx} = 3(2x-1)^2 \times 2 = 6$ at $x = 1$. $y - 1 = 6(x-1)$, so $y = 6x - 5$. *cf. 2011 HSC Q2(c)*

**32.** $\dfrac{dy}{dx} = 2x - 3 = -1$ at $x = 1$, so the normal has gradient 1. $y + 2 = x - 1$, so $x - y - 3 = 0$.

**33.** $f(x)f(-x) = (e^x - 1)(e^{-x} - 1) = 1 - e^x - e^{-x} + 1 = 2 - e^x - e^{-x}$. $f(x) + f(-x) = e^x + e^{-x} - 2$. The sum is 0. *cf. 2010 HSC Q4(d)*

**34.** $x = e^3 = 20.0855\ldots \approx 20.0855$ *cf. 2009 HSC Q1(f)*

**35.** $2^{2x-1} = 2^5$, so $2x - 1 = 5$ and $x = 3$. *cf. 2011 HSC Q1(c)*

**36.** Let $u = \ln x$: $u^2 - u - 2 = 0$, so $(u-2)(u+1) = 0$. $\ln x = 2$ or $\ln x = -1$, so $x = e^2$ or $x = e^{-1}$. *cf. 2008 HSC Q7(a)*

**37.** $\log_a 12 = 2\log_a 2 + \log_a 3 = 0.72 + 0.57 = 1.29$

**38.** $3\ln x - \frac{1}{2}\ln(x+1)$

**39.** $\sin\theta = \dfrac{\sqrt{3}}{2}$, related angle $\dfrac{\pi}{3}$; in the second quadrant $\theta = \dfrac{2\pi}{3}$. *cf. 2009 HSC Q1(e)*

**40.** $\cos x = -\dfrac{1}{\sqrt{2}}$, related angle $\dfrac{\pi}{4}$, second and third quadrants: $x = \dfrac{3\pi}{4}$ or $\dfrac{5\pi}{4}$. *cf. 2011 HSC Q2(b)*

**41.** $\sin\dfrac{x}{2} = \pm\dfrac{1}{\sqrt{2}}$ with $-\dfrac{\pi}{2} \le \dfrac{x}{2} \le \dfrac{\pi}{2}$, so $\dfrac{x}{2} = \pm\dfrac{\pi}{4}$ and $x = \pm\dfrac{\pi}{2}$. *cf. 2008 HSC Q6(a)*

**42.** $1 - \cos^2\theta = \sin^2\theta$, so the left side is $\dfrac{\sin^2\theta}{\sin\theta\cos\theta} = \dfrac{\sin\theta}{\cos\theta} = \tan\theta$.

**43.** Need $x + 2 > 0$: $x > -2$, that is $(-2, \infty)$. *cf. 2010 HSC Q1(g)*

**44.** $f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 5$. $2x^2 - 5 = 7$ gives $x^2 = 6$, so $x = \pm\sqrt{6}$.

**45.** $f(-x) = -x^3 + 4x = -(x^3 - 4x) = -f(x)$, so $f$ is odd. *cf. 2008 HSC Q8(a)(ii)*

**46.** $(x - 5)(x + 3) < 0$, so $-3 < x < 5$. *cf. 2010 HSC Q2(b)*

**47.** $x = 3y - 4$, so $f^{-1}(x) = \dfrac{x + 4}{3}$.

**48.** Translate right 1, dilate vertically by a factor of 2, then translate up 3.

**49.** $1 - P(\text{neither}) = 1 - \dfrac{4}{5} \times \dfrac{3}{4} = 1 - \dfrac{3}{5} = \dfrac{2}{5}$ *cf. 2009 HSC Q9(a)*

**50.** (a) $\dfrac{4}{10} \times \dfrac{3}{9} = \dfrac{2}{15}$ (b) $2 \times \dfrac{4}{10} \times \dfrac{6}{9} = \dfrac{8}{15}$ *cf. 2010 HSC Q4(c)*

**51.** $-7 \le 2x - 1 \le 7$, so $-6 \le 2x \le 8$ and $-3 \le x \le 4$. *cf. 2011 HSC Q1(e)*

**52.** $\dfrac{dy}{dx} = 2x = 1$, so $\tan\theta = 1$ and $\theta = 45°$.

**53.** a) $e^{x\ln 9}$ b) $e^{5x\ln 2}$

**54.** a) $x^2e^{2x}(3 + 2x)$ b) $e^{3x}(1 - 3x)$

**55.** a) $(\ln 6)\,6^x$ b) $5(\ln 2)\,2^x$ c) $3^{4x} = e^{4x\ln 3}$, so $\dfrac{dy}{dx} = 4(\ln 3)\,3^{4x}$

**56.** a) $e^x + xe^x = e^x(1 + x)$ b) $2xe^{4x} + 4x^2e^{4x} = 2xe^{4x}(1 + 2x)$

**57.** $\dfrac{dy}{dx} = 4x^3e^x + x^4e^x = x^3e^x(4 + x)$

**58.** $\dfrac{dy}{dx} = 3e^{-x} - (3x - 1)e^{-x} = e^{-x}(4 - 3x)$

**59.** $\dfrac{dy}{dx} = e^{-2x} - 2xe^{-2x} = e^{-2x}(1 - 2x)$

**60.** $\dfrac{dy}{dx} = 2xe^{-3x} - 3x^2e^{-3x} = xe^{-3x}(2 - 3x)$

**61.** $\dfrac{dy}{dx} = \dfrac{3e^{3x} \cdot x - e^{3x} \cdot 1}{x^2} = \dfrac{e^{3x}(3x - 1)}{x^2}$

**62.** $\dfrac{dy}{dx} = \dfrac{1 \cdot e^{2x} - x \cdot 2e^{2x}}{(e^{2x})^2} = \dfrac{e^{2x}(1 - 2x)}{e^{4x}} = \dfrac{1 - 2x}{e^{2x}}$

**63.** $\dfrac{dy}{dx} = \dfrac{e^x(x+1) - e^x \cdot 1}{(x+1)^2} = \dfrac{xe^x}{(x+1)^2}$

**64.** $\dfrac{dy}{dx} = 2^x + x(\ln 2)\, 2^x = 2^x(1 + x \ln 2)$

**65.** $\dfrac{dy}{dx} = 2x \cdot 5^x + x^2(\ln 5)\, 5^x = x \cdot 5^x(2 + x \ln 5)$

**66.** $\dfrac{dy}{dx} = (\ln 2)\, 2^x e^x + 2^x e^x = 2^x e^x(\ln 2 + 1)$

**67.** $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 2) - e^x \cdot e^x}{(e^x + 2)^2} = \dfrac{2e^x}{(e^x + 2)^2}$

**68.** $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 1) - (e^x - 1)e^x}{(e^x + 1)^2} = \dfrac{2e^x}{(e^x + 1)^2}$

**69.** $\dfrac{dy}{dx} = \dfrac{2e^{2x}(1 + e^{2x}) - e^{2x} \cdot 2e^{2x}}{(1 + e^{2x})^2} = \dfrac{2e^{2x}}{(1 + e^{2x})^2}$

**70.** $\dfrac{dy}{dx} = e^{-x}(1 - x)$. $e^{-x} > 0$, so $x = 1$ and $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**71.** $\dfrac{dy}{dx} = xe^x(x + 2)$. $e^x > 0$, so $x = 0$ or $x = -2$, giving $(0,\, 0)$ and $\left(-2,\, 4e^{-2}\right)$.

**72.** $\dfrac{dy}{dx} = \dfrac{e^x(x - 1)}{x^2}$, which is 0 when $x = 1$. $y = e$. The point is $(1,\, e)$.

**73.** $a = e^{\ln a}$, so $a^x = e^{x \ln a}$. Since $\ln a$ is a constant, $\dfrac{d}{dx}\left(e^{x \ln a}\right) = \ln a \cdot e^{x \ln a} = (\ln a)\, a^x$.

**74.** The student used the power rule, which only applies when $x$ is the base. Here $x$ is the power, so $\dfrac{d}{dx}\left(2^x\right) = (\ln 2)\, 2^x$.

**75.** $\dfrac{d}{dx}\left(xe^x\right) = e^x + xe^x = (x + 1)e^x$. At $x = 0$: $(0 + 1)e^0 = 1$.

**76.** $\dfrac{dy}{dx} = e^{2x}(1 + 2x)$. At $x = 1$: $e^2(1 + 2) = 3e^2$.

**77.** a) $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$ b) For $x < 1$, $1 - x > 0$ and $e^{-x} > 0$, so $f'(x) > 0$. c) $f'(x) = 0$ when $x = 1$; $f(1) = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**78.** $3(e^x + 2)^2 \times e^x = 3e^x(e^x + 2)^2$ *cf. 2009 HSC Q2(a)(ii)*

**79.** $-10e^{-2x}$

**80.** $f'(x) = (2x - 2)e^{x^2 - 2x}$, so $f'(2) = 2e^0 = 2$.

**81.** $\dfrac{dy}{dx} = 6e^{2x} = 2 \times 3e^{2x} = 2y$

**82.** $\dfrac{dy}{dx} = (2x + k)e^{x^2 + kx}$. $e^{x^2 + kx} > 0$, so $2(-2) + k = 0$ and $k = 4$.

**83.** Let $u = e^x$: $u^2 - 5u + 6 = 0$, so $u = 2$ or $u = 3$. $x = \ln 2$ or $x = \ln 3$. *cf. 2008 HSC Q7(a)*

**84.** $x = \dfrac{\ln 20}{\ln 3} = 2.726\ldots \approx 2.73$

**85.** $\dfrac{\ln 40}{\ln 5} = 2.292\ldots \approx 2.292$

**86.** $\ln 40 = \ln(2^3 \times 5) = 3a + b$; $\quad \ln 0.4 = \ln \dfrac{2}{5} = a - b$

**87.** $\dfrac{1}{2}(16 - x^2)^{-\frac{1}{2}} \times (-2x) = -\dfrac{x}{\sqrt{16 - x^2}}$ *cf. 2011 HSC Q4(d)(i)*

**88.** $\dfrac{(2x-1) - 2(x+1)}{(2x-1)^2} = -\dfrac{3}{(2x-1)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**89.** $\dfrac{dy}{dx} = 3x^2 - 4x = 4$ at $x = 2$. $y - 1 = 4(x - 2)$, so $y = 4x - 7$.

**90.** $\dfrac{dy}{dx} = 2x = -4$. $y - 4 = -4(x + 2)$, so $y = -4x - 4$. *cf. 2010 HSC Q7(b)(i)*

**91.** $\Delta = (k+2)^2 - 4(k+5) = k^2 - 16 = 0$, so $k = \pm 4$. *cf. 2009 HSC Q4(b)*

**92.** $x(x - 5) = 0$, so $x = 0$ or $x = 5$. *cf. 2010 HSC Q1(a)*

**93.** $\tan x = \dfrac{1}{\sqrt{3}}$: $x = \dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$.

**94.** $\cos x = \pm\dfrac{1}{\sqrt{2}}$: $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$. *cf. 2008 HSC Q6(a)*

**95.** (a) $\theta = \dfrac{\ell}{r} = \dfrac{4\pi}{6} = \dfrac{2\pi}{3}$ (b) $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 36 \times \dfrac{2\pi}{3} = 12\pi$ cm$^2$ *cf. 2010 HSC Q6(b)*

**96.** $2x - 6 > 0$, so $x > 3$. *cf. 2010 HSC Q1(g)*

**97.** $f(g(x)) = (x + 3)^2$, with range $y \geq 0$.

**98.** $f(-x) = (-x)^4 - 5(-x)^2 = x^4 - 5x^2 = f(x)$ *cf. 2008 HSC Q8(a)(ii)*

**99.** $-9 \leq 3 - 2x \leq 9$, so $-12 \leq -2x \leq 6$ and $-3 \leq x \leq 6$. *cf. 2011 HSC Q1(e)*

**100.** $\dfrac{3}{5} \times \dfrac{2}{4} + \dfrac{2}{5} \times \dfrac{1}{4} = \dfrac{6}{20} + \dfrac{2}{20} = \dfrac{2}{5}$ *cf. 2011 HSC Q5(b)*

**101.** $M = (1, 4)$. $y - 4 = -\frac{1}{2}(x - 1)$, so $x + 2y - 9 = 0$. *cf. 2008 HSC Q2(b)*

**102.** $\dfrac{3(\sqrt{7}+2)}{7-4} = \sqrt{7} + 2$, so $a = 2$, $b = 1$. *cf. 2010 HSC Q1(b)*

**103.** a) $e^4$ b) $3e^0 = 3$ c) $e^3$

**104.** a) $-\dfrac{1}{5e}$ b) $-\dfrac{1}{e^4}$ c) $\dfrac{e}{3}$

**105.** Point $(2,\, e^2)$, $m = e^2$. $y - e^2 = e^2(x-2)$, so $y = e^2x - e^2$.

**106.** Point $(0,\, 1)$, $m = 5e^0 = 5$. $y - 1 = 5x$, so $y = 5x + 1$.

**107.** Point $(1,\, 2e^3)$, $m = 6e^3$. $y - 2e^3 = 6e^3(x-1)$, so $y = 6e^3x - 4e^3$.

**108.** Point $(0,\, 1)$. Tangent gradient 3, normal gradient $-\frac{1}{3}$. $y - 1 = -\frac{1}{3}x$, so $x + 3y - 3 = 0$.

**109.** Point $(1,\, e)$. Tangent gradient $e$, normal gradient $-\dfrac{1}{e}$. $e(y-e) = -(x-1)$, so $x + ey - e^2 - 1 = 0$.

**110.** Point $(0,\, 1)$. Tangent gradient $-2$, normal gradient $\frac{1}{2}$. $y - 1 = \frac{1}{2}x$, so $x - 2y + 2 = 0$.

**111.** $\dfrac{dy}{dx} = e^{2x}(1+2x)$, so $m = 1$ at $x = 0$. The tangent is $y = x$.

**112.** Point $(2,\, 2e^{-2})$. $\dfrac{dy}{dx} = e^{-x}(1-x)$, so $m = -e^{-2}$. $y - 2e^{-2} = -e^{-2}(x-2)$, so $y = -e^{-2}x + 4e^{-2}$.

**113.** Point $(1,\, e)$. $\dfrac{dy}{dx} = xe^x(2+x)$, so $m = 3e$. $y - e = 3e(x-1)$, so $y = 3ex - 2e$.

**114.** Point $(-1,\, e)$. $\dfrac{dy}{dx} = 2xe^{x^2} = -2e$, so the normal gradient is $\dfrac{1}{2e}$. $2e(y-e) = x+1$, so $x - 2ey + 2e^2 + 1 = 0$.

**115.** Point $(1,\, 1)$. Tangent gradient $2e^0 = 2$, normal gradient $-\frac{1}{2}$. $y - 1 = -\frac{1}{2}(x-1)$, so $x + 2y - 3 = 0$.

**116.** Point $(2,\, 1)$. Tangent gradient $2(2)e^0 = 4$, normal gradient $-\frac{1}{4}$. $y - 1 = -\frac{1}{4}(x-2)$, so $x + 4y - 6 = 0$.

**117.** Point $(a,\, e^{2a})$; $\dfrac{dy}{dx} = 2e^{2x}$, so $m = 2e^{2a}$. Tangent: $y - e^{2a} = 2e^{2a}(x-a)$. At $y = 0$: $-e^{2a} = 2e^{2a}(x-a)$. Since $e^{2a} \neq 0$, $x - a = -\frac{1}{2}$, so $x = a - \frac{1}{2}$.

**118.** Point $(a,\, e^{-a})$; $\dfrac{dy}{dx} = -e^{-x}$, so $m = -e^{-a}$. Tangent: $y - e^{-a} = -e^{-a}(x-a)$. At $y = 0$: $-e^{-a} = -e^{-a}(x-a)$. Since $e^{-a} \neq 0$, $x - a = 1$, so $x = a + 1$.

**119.** Tangent: $y - e^a = e^a(x-a)$. At $x = 0$: $y = e^a - ae^a = (1-a)e^a$.

**120.** Tangent: $y - e^{2a} = 2e^{2a}(x-a)$. Through $(0,\, 0)$: $-e^{2a} = -2ae^{2a}$, and $e^{2a} > 0$, so $a = \frac{1}{2}$. The point is $\left(\frac{1}{2},\, e\right)$ with $m = 2e$, so the tangent is $y = 2ex$.

**121.** $e^x = 3$, so $x = \ln 3$ and $y = 3$. The point is $(\ln 3,\, 3)$.

**122.** The tangent gradient is 6: $3e^{3x} = 6$, so $e^{3x} = 2$ and $x = \frac{1}{3}\ln 2$. Then $y = 2$. The point is $\left(\frac{1}{3}\ln 2,\, 2\right)$.

**123.** A horizontal tangent has gradient 0. The gradient is $\dfrac{dy}{dx} = e^x$, which is greater than 0 for every $x$, so it is never 0.

**124.** For $y = e^x$: $m_1 = e^0 = 1$. For $y = e^{-x}$: $m_2 = -e^0 = -1$. $m_1 m_2 = -1$, so the tangents are perpendicular.

**125.** Point $(0,\, 1)$; tangent gradient 1, normal gradient $-1$. Normal: $y = -x + 1$, which crosses at $(1,\, 0)$.

**126.** The $y$-value must come from the original function, $y = e^2$, not from the derivative $2e^2$. The correct point is $(1,\, e^2)$.

**127.** a) Point $(0,\, 1)$, $f'(x) = 2e^{2x}$ so $m = 2$. Tangent: $y = 2x + 1$. b) $2x + 1 = 0$, so $x = -\frac{1}{2}$: the point $\left(-\frac{1}{2},\, 0\right)$.

**128.** $2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$ *cf. 2011 HSC Q2(d)*

**129.** $\dfrac{2e^{2x}(x+1) - e^{2x}}{(x+1)^2} = \dfrac{e^{2x}(2x+1)}{(x+1)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**130.** $2(\ln 3)\, 3^{2x}$

**131.** $2(1 - e^x) \times (-e^x) = -2e^x(1 - e^x)$ *cf. 2009 HSC Q2(a)(ii)*

**132.** $f'(x) = e^{-2x} + x \times (-2e^{-2x}) = e^{-2x}(1 - 2x)$

**133.** $x = 0$: $a = 4$. $\dfrac{dy}{dx} = abe^{bx}$, so $ab = -8$ and $b = -2$.

**134.** $3^{2x+1} = 3^4$, so $2x + 1 = 4$ and $x = \frac{3}{2}$. *cf. 2011 HSC Q1(c)*

**135.** $x = \dfrac{\ln 10}{\ln 2} = 3.3219\ldots \approx 3.32$

**136.** Let $u = 2^x$: $u^2 - 3u - 4 = 0$, so $(u - 4)(u + 1) = 0$. $2^x = 4$ gives $x = 2$; $2^x = -1$ has no solution since $2^x > 0$. *cf. 2008 HSC Q7(a)*

**137.** (a) $B(1,\, 0)$; gradient of $BC$ is $-\frac{1}{3}$: $y = -\frac{1}{3}(x - 1)$, i.e. $x + 3y - 1 = 0$. (b) $A(0,\, -3)$, $C\left(0,\, \frac{1}{3}\right)$, so $AC = \frac{10}{3}$ and the height from $B$ is 1: area $= \frac{1}{2} \times \frac{10}{3} \times 1 = \frac{5}{3}$ square units. *cf. 2009 HSC Q5(a)*

**138.** Gradients $-\frac{2}{5}$ and $\frac{5}{2}$; their product is $-1$, so the lines are perpendicular. *cf. 2011 HSC Q3(c)(ii)*

**139.** $\sqrt{4^2 + 3^2} = 5$ *cf. 2008 HSC Q3(a)(iii)*

**140.** $(2x + 3)^{-2}$: $-2(2x + 3)^{-3} \times 2 = -\dfrac{4}{(2x+3)^3}$

**141.** $\dfrac{2x(2x+1) - 2x^2}{(2x+1)^2} = \dfrac{2x(x+1)}{(2x+1)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**142.** $3x + 4 = 5x$, so $x = 2$. *cf. 2009 HSC Q1(b)*

**143.** $x = \dfrac{-2 \pm \sqrt{4+8}}{2} = -1 \pm \sqrt{3}$

**144.** Related angle $\dfrac{\pi}{6}$, second quadrant: $\theta = \dfrac{5\pi}{6}$. *cf. 2009 HSC Q1(e)*

**145.** $x = \dfrac{7\pi}{6}$ or $\dfrac{11\pi}{6}$ *cf. 2011 HSC Q2(b)*

**146.** $y > 2$

**147.** $y$-intercept $(0, 2)$; asymptote $y = 3$ as $x \to \infty$; the curve increases towards it (reflection of $y = e^{-x}$ in the $x$-axis, moved up 3). *cf. 2010 HSC Q3(b)(i)*

**148.** $x = e^y + 1$, so $f^{-1}(x) = \ln(x-1)$, with domain $x > 1$.

**149.** $x - 2 = 7$ or $x - 2 = -7$: $x = 9$ or $x = -5$. *cf. 2009 HSC Q1(c)*

**150.** (a) $0.2 \times 0.2 = 0.04$ (b) $0.8 \times 0.2 = 0.16$ *cf. 2008 HSC Q9(a)*

**151.** $15 + \sqrt{5} - 6\sqrt{5} - 2 = 13 - 5\sqrt{5}$ *cf. 2008 HSC Q1(e)*

**152.** $\dfrac{4x^6}{4x^2} = x^4$

**153.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 7$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

**154.** a) $3\ln x + \ln(x+2)$ b) $\ln(x-1) - \ln(x+5)$ c) $2\ln(4x+3)$

**155.** a) Yes (product) b) No (a sum) c) Yes (power)

**156.** a) $\dfrac{2x}{x^2+1}$ b) $\dfrac{3x^2-4}{x^3-4x}$

**157.** $\dfrac{dy}{dx} = \dfrac{4x-5}{2x^2-5x+3}$

**158.** $\dfrac{dy}{dx} = \dfrac{6}{6x} = \dfrac{1}{x}$

**159.** $\dfrac{dy}{dx} = \dfrac{3}{3x+2}$

**160.** $\dfrac{dy}{dx} = \dfrac{-4}{1-4x} = \dfrac{4}{4x-1}$

**161.** $\dfrac{dy}{dx} = 5 \times \dfrac{2}{2x+3} = \dfrac{10}{2x+3}$

**162.** $\dfrac{dy}{dx} = 3x^2 - \dfrac{2}{x}$

**163.** $\dfrac{dy}{dx} = \dfrac{2x}{x^2+1} + 2e^{2x}$

**164.** $y = 3\ln(2x+1)$, so $\dfrac{dy}{dx} = \dfrac{6}{2x+1}$.

**165.** $y = \frac{1}{2}\ln(4x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2} \times \dfrac{4}{4x-1} = \dfrac{2}{4x-1}$.

**166.** $y = 5\ln x$, so $\dfrac{dy}{dx} = \dfrac{5}{x}$.

**167.** $y = \ln x + \ln(x+3)$, so $\dfrac{dy}{dx} = \dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{2x+3}{x(x+3)}$.

**168.** $\dfrac{dy}{dx} = \dfrac{1}{x+2} - \dfrac{1}{x-1} = \dfrac{(x-1)-(x+2)}{(x+2)(x-1)} = \dfrac{-3}{(x+2)(x-1)}$

**169.** $\dfrac{dy}{dx} = \dfrac{3}{3x-1} - \dfrac{1}{x+1} = \dfrac{3(x+1)-(3x-1)}{(3x-1)(x+1)} = \dfrac{4}{(3x-1)(x+1)}$

**170.** $y = 3\ln x + \frac{1}{2}\ln(x+1)$, so $\dfrac{dy}{dx} = \dfrac{3}{x} + \dfrac{1}{2(x+1)} = \dfrac{7x+6}{2x(x+1)}$.

**171.** $y = 2\ln(x+1) - \ln(x-2)$, so $\dfrac{dy}{dx} = \dfrac{2}{x+1} - \dfrac{1}{x-2} = \dfrac{x-5}{(x+1)(x-2)}$.

**172.** $y = \frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(\dfrac{1}{x+1} - \dfrac{1}{x-1}\right) = \frac{1}{2} \times \dfrac{-2}{x^2-1} = -\dfrac{1}{x^2-1}$.

**173.** Differentiate both sides with the chain rule: $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$. Since $e^{\ln x} = x$, this gives $x\dfrac{d}{dx}(\ln x) = 1$, so $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

**174.** $\dfrac{d}{dx}\ln(5x) = \dfrac{5}{5x} = \dfrac{1}{x}$. By the log laws $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant with derivative 0.

**175.** The student forgot to multiply by the derivative of the inside. Correct: $\dfrac{2x}{x^2+1}$.

**176.** $\dfrac{dy}{dx} = \dfrac{2x}{x^2+3} = \dfrac{2}{4} = \dfrac{1}{2}$ at $x = 1$.

**177.** a) $f'(x) = \dfrac{2x-2}{x^2-2x+5}$ b) $x^2 - 2x + 5 = (x-1)^2 + 4 > 0$ for all $x$, so the log is always defined. c) $f'(x) = 0$ when $x = 1$; $f(1) = \ln 4$. The point is $(1, \ln 4)$.

**178.** $3e^{3x} - e^{-x}$

**179.** $2xe^{2x} + 2(x^2+1)e^{2x} = 2e^{2x}(x^2+x+1)$ *cf. 2011 HSC Q2(d)*

**180.** $3(\ln 5)\,5^x$

**181.** Point $(0, 1)$, $m = 2e^0 = 2$: $y = 2x + 1$.

**182.** Tangent gradient 1, normal gradient $-1$: $y = -x + 1$.

**183.** Point $(1, e)$, $m = e$: $y - e = e(x-1)$, so $y = ex$, which passes through $(0, 0)$.

**184.** $e^{\ln 9} = 9$

**185.** $x = e^{1.5} = 4.4816\ldots \approx 4.4817$ *cf. 2009 HSC Q1(f)*

**186.** $x + 1 = 2^3 = 8$, so $x = 7$.

**187.** $\ln 9 + \ln 4 = \ln 36$

**188.** $e^{0.2t} = 8$, so $0.2t = \ln 8$ and $t = 5\ln 8 = 15\ln 2$.

**189.** $\log_4 x = \dfrac{\log_2 x}{\log_2 4} = \dfrac{\log_2 x}{2}$

**190.** $5(3x^2-1)^4 \times 6x = 30x(3x^2-1)^4$ *cf. 2008 HSC Q2(a)(i)*

**191.** $x^{\frac{3}{2}}$: $\frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$

**192.** $2x-6=4$, so $x=5$ and $y=-5$: $(5,\,-5)$. *cf. 2010 HSC Q7(b)(ii)*

**193.** $f'(x) = 3x^2+6x+k \geq 0$ for all $x$: need $\Delta = 36-12k \leq 0$, so $k \geq 3$. *cf. 2010 HSC Q8(d)*

**194.** $\Delta = 1-12 = -11 < 0$, so there are no real roots.

**195.** $\frac{1}{2} \times 2 \times 2\sin\theta = \sqrt{3}$, so $\sin\theta = \dfrac{\sqrt{3}}{2}$ and the other value is $\theta = \dfrac{2\pi}{3}$. *cf. 2009 HSC Q5(c)(i)*

**196.** $\cos x = \dfrac{\sqrt{3}}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{11\pi}{6}$ *cf. 2011 HSC Q2(b)*

**197.** $4-x \geq 0$, so $x \leq 4$. *cf. 2010 HSC Q1(g)*

**198.** $x$-intercept $(1,\,0)$; vertical asymptote $x=0$; increasing, passing through $(e,\,1)$. *cf. 2010 HSC Q3(b)(i)*

**199.** $2x+3=9$ or $2x+3=-9$: $x=3$ or $x=-6$. *cf. 2010 HSC Q1(d)*

**200.** $f^{-1}(x) = \dfrac{x+1}{2}$, so $f^{-1}(5) = 3$.

**201.** $1-\dfrac{7}{8}\times\dfrac{9}{10} = 1-\dfrac{63}{80} = \dfrac{17}{80}$ *cf. 2009 HSC Q9(a)*

**202.** $\dfrac{x(x-4)}{(x-4)(x+4)} = \dfrac{x}{x+4}$

**203.** a) $\dfrac{\ln x}{\ln 6}$ b) $\dfrac{\ln\left(x^2-1\right)}{\ln 3}$

**204.** a) $\dfrac{1}{x\ln 4}$ b) $\dfrac{3}{x\ln 5}$ c) $\dfrac{3}{(3x-1)\ln 2}$

**205.** a) Log law b) Chain rule c) Chain rule d) Log law

**206.** $\dfrac{dy}{dx} = 1\times\ln x + x\times\dfrac{1}{x} = \ln x+1$

**207.** $\dfrac{dy}{dx} = 5x^4\ln x + x^4 = x^4(5\ln x+1)$

**208.** $\dfrac{dy}{dx} = e^x\ln x + \dfrac{e^x}{x} = e^x\left(\ln x+\dfrac{1}{x}\right)$

**209.** $\dfrac{dy}{dx} = 2x\ln(3x) + x^2\times\dfrac{1}{x} = x\big(2\ln(3x)+1\big)$

**210.** $\dfrac{dy}{dx} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{\sqrt{x}}{x} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{2}{2\sqrt{x}} = \dfrac{\ln x+2}{2\sqrt{x}}$

**211.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}\cdot x^2 - \ln x\cdot 2x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$

**212.** $\dfrac{dy}{dx} = \dfrac{1 \cdot \ln x - x \cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{\ln x - 1}{(\ln x)^2}$

**213.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}(x+1) - \ln x}{(x+1)^2} = \dfrac{x + 1 - x\ln x}{x(x+1)^2}$

**214.** $\dfrac{dy}{dx} = 2\ln x \times \dfrac{1}{x} = \dfrac{2\ln x}{x}$

**215.** $\dfrac{dy}{dx} = 4(\ln x)^3 \times \dfrac{1}{x} = \dfrac{4(\ln x)^3}{x}$

**216.** $y = (\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{\ln x}}$.

**217.** Point $(1, 0)$, $m = 1$. The tangent is $y = x - 1$.

**218.** Point $\left(\frac{1}{3}, \ln 1\right) = \left(\frac{1}{3}, 0\right)$. $\dfrac{dy}{dx} = \dfrac{1}{x}$, so $m = 3$. $y = 3\left(x - \frac{1}{3}\right) = 3x - 1$.

**219.** Point $(e, 2)$. $\dfrac{dy}{dx} = \dfrac{2}{x}$, so $m = \dfrac{2}{e}$. $y - 2 = \dfrac{2}{e}(x - e)$, so $y = \dfrac{2x}{e}$.

**220.** Point $(e, 1)$, tangent gradient $\dfrac{1}{e}$, normal gradient $-e$. $y - 1 = -e(x - e)$, so $y = -ex + e^2 + 1$.

**221.** Point $(1, 0)$. $\dfrac{dy}{dx} = x(2\ln x + 1)$, so $m = 1$. The tangent is $y = x - 1$.

**222.** Point $(1, 0)$. $\dfrac{dy}{dx} = \ln x + 1 = 1$, so the normal gradient is $-1$. The normal is $y = -x + 1$.

**223.** By the change of base law, $\log_a x = \dfrac{\ln x}{\ln a} = \dfrac{1}{\ln a} \cdot \ln x$. Since $\dfrac{1}{\ln a}$ is a constant, $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{\ln a} \cdot \dfrac{1}{x} = \dfrac{1}{x \ln a}$.

**224.** $\ln(x^2) = 2\ln x$, with derivative $\dfrac{2}{x}$. $(\ln x)^2$ needs the chain rule: $\dfrac{2\ln x}{x}$. In the first the power is inside the log; in the second it is on the whole log.

**225.** Point $(e, 1)$, $m = \dfrac{1}{e}$. $y - 1 = \dfrac{1}{e}(x - e)$ gives $y = \dfrac{x}{e}$, which passes through $(0, 0)$.

**226.** $\dfrac{dy}{dx} = \ln x + 1 = 0$, so $\ln x = -1$ and $x = e^{-1}$. $y = e^{-1}\ln e^{-1} = -e^{-1}$. The point is $(e^{-1}, -e^{-1})$.

**227.** a) $f'(x) = \dfrac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$ b) $1 - \ln x = 0$, so $x = e$; $f(e) = \dfrac{1}{e}$. The point is $\left(e, \dfrac{1}{e}\right)$.

**228.** $\dfrac{2x}{x^2 + 7}$

**229.** $\dfrac{1}{x+1} - \dfrac{1}{x-1} = \dfrac{(x-1) - (x+1)}{(x+1)(x-1)} = -\dfrac{2}{x^2 - 1}$

**230.** $e^{-x} - xe^{-x} = e^{-x}(1 - x)$ *cf. 2011 HSC Q2(d)*

**231.** $(\ln 10)\,10^x$

**232.** $\dfrac{1}{2\sqrt{x}}e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$

**233.** $\dfrac{dy}{dx} = \dfrac{4}{4x} = \dfrac{1}{x}$, so the gradient is $\dfrac{1}{3}$. *cf. 2010 HSC Q2(c)*

**234.** $2e^{2x} - (-2e^{-2x}) = 2e^{2x} + 2e^{-2x} = 2(e^{2x} + e^{-2x})$ *cf. 2010 HSC Q4(d)*

**235.** $(e^x - 1)(e^x - 2) = 0$, so $x = 0$ or $x = \ln 2$. *cf. 2008 HSC Q7(a)*

**236.** $\dfrac{3}{3x-4}$ *cf. 2011 HSC Q1(d)*

**237.** $x + x = 2x$

**238.** $4^{\frac{1}{2}} = 2$, so $\log_4 2 = \frac{1}{2}$.

**239.** $\ln \dfrac{18 \times 5}{2} = \ln 45$

**240.** Let $u = \log_2 x$: $u^2 - 4u + 3 = 0$, so $u = 1$ or $u = 3$, giving $x = 2$ or $x = 8$. *cf. 2008 HSC Q7(a)*

**241.** $6x^{-\frac{1}{2}}$: $-3x^{-\frac{3}{2}} = -\dfrac{3}{x\sqrt{x}}$

**242.** Tangent gradient 2, normal gradient $-\frac{1}{2}$: $y - 1 = -\frac{1}{2}(x - 1)$, so $x + 2y - 3 = 0$.

**243.** $x(x-2)(x+2) = 0$: $x = 0,\ \pm 2$.

**244.** $\Delta = k^2 - 36 = 0$, so $k = \pm 6$. *cf. 2009 HSC Q4(b)*

**245.** Related angle $\dfrac{\pi}{3}$, third and fourth quadrants: $x = \dfrac{4\pi}{3}$ or $\dfrac{5\pi}{3}$. *cf. 2011 HSC Q2(b)*

**246.** $\sin x = \pm\dfrac{1}{\sqrt{2}}$: $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$. *cf. 2008 HSC Q6(a)*

**247.** $A = 3$; $\dfrac{2\pi}{b} = \pi$, so $b = 2$. *cf. 2010 HSC Q8(c)*

**248.** The region between the parabola $y = 9 - x^2$ and the $x$-axis, for $-3 \leq x \leq 3$, including the boundary. *cf. 2009 HSC Q3(c)*

**249.** $x + 1 \geq 3$ or $x + 1 \leq -3$: $x \geq 2$ or $x \leq -4$. *cf. 2011 HSC Q1(e)*

**250.** $g(4) = 25$, so $f(25) = 5$.

**251.** (a) $3 \times \dfrac{4}{12} \times \dfrac{3}{11} = \dfrac{3}{11}$ (b) $1 - \dfrac{3}{11} = \dfrac{8}{11}$ *cf. 2010 HSC Q4(c)*

**252.** $(x-2)(x^2 + 2x + 4)$

**253.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 1$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

**254.** a) $\cos x - \sin x$ b) $4\sec^2 x$ c) $2x + 3\sin x$

**255.** $\dfrac{dy}{dx} = (2x+3)\cos\left(x^2+3x\right)$

**256.** $\dfrac{dy}{dx} = 3x^2\cos\left(x^3\right)$

**257.** $\dfrac{dy}{dx} = 5\cos(5x-2)$

**258.** $\dfrac{dy}{dx} = -2x\sin\left(x^2+1\right)$

**259.** $\dfrac{dy}{dx} = -4\sin 4x$

**260.** $\dfrac{dy}{dx} = -(-3)\sin(2-3x) = 3\sin(2-3x)$

**261.** $\dfrac{dy}{dx} = 6\sec^2 6x$

**262.** $\dfrac{dy}{dx} = 2x\sec^2\left(x^2\right)$

**263.** $\dfrac{dy}{dx} = -\sec^2(1-x)$

**264.** $\dfrac{dy}{dx} = 15\cos 3x - 8\sin 4x$

**265.** $\dfrac{dy}{dx} = 2 \times \frac{1}{2}\sec^2\dfrac{x}{2} = \sec^2\dfrac{x}{2}$

**266.** $\dfrac{dy}{dx} = -\pi\sin(\pi x) - \pi\cos(\pi x)$

**267.** $\dfrac{dy}{dx} = -2\sin 2x$. At $x = \dfrac{\pi}{12}$: $-2\sin\dfrac{\pi}{6} = -1$.

**268.** $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. At $x = \dfrac{\pi}{4}$: $\dfrac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = 2$.

**269.** $\dfrac{dy}{dx} = 3\cos 3x$. At $x = \dfrac{\pi}{9}$: $3\cos\dfrac{\pi}{3} = \dfrac{3}{2}$.

**270.** $\dfrac{dy}{dx} = \cos x + \sin x = 0$, so $\tan x = -1$: $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$.

**271.** $\dfrac{dy}{dx} = 1 - 2\sin x = 0$, so $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

**272.** $\dfrac{dy}{dx} = 2\cos 2x = 0$, so $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$: $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

**273.** $\tan x = \dfrac{\sin x}{\cos x}$, so $\dfrac{d}{dx}(\tan x) = \dfrac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x} = \sec^2 x$.

**274.** At $x = 0$ the curve rises most steeply, with gradient 1; at $x = \frac{\pi}{2}$ it reaches a maximum, so the tangent is horizontal; at $x = \pi$ it falls most steeply, with gradient $-1$. These are the values of $\cos x$ at 0, $\frac{\pi}{2}$ and $\pi$.

**275.** The student forgot to multiply by the derivative of the angle, 3. Correct: $-3\sin 3x$.

**276.** $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. Since $0 < \cos^2 x \leq 1$ wherever $\tan x$ is defined, the gradient is always at least 1, so it is never 0.

**277.** a) $f'(x) = \cos x - \sqrt{3}\sin x$ b) $\cos x = \sqrt{3}\sin x$, so $\tan x = \dfrac{1}{\sqrt{3}}$: $x = \dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$.

**278.** $\dfrac{5}{5x-2}$ *cf. 2011 HSC Q1(d)*

**279.** $3x^2 \ln x + x^3 \times \dfrac{1}{x} = x^2(3\ln x + 1)$ *cf. 2008 HSC Q2(a)(ii)*

**280.** $\dfrac{1}{x \ln 3}$

**281.** $2\ln x \times \dfrac{1}{x} = \dfrac{2\ln x}{x}$

**282.** $-3e^{-3x}$

**283.** Product rule: $\ln(2+x) + (2+x) \times \dfrac{1}{2+x} - 1 = \ln(2+x) + 1 - 1 = \ln(2+x)$ *cf. 2009 HSC Q10(e)*

**284.** Point $(e^2, 2)$, $m = \dfrac{1}{e^2}$: $y - 2 = \dfrac{1}{e^2}(x - e^2)$, so $y = \dfrac{x}{e^2} + 1$.

**285.** $-\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}} = 0$

**286.** $\theta = \dfrac{\ell}{r} = \dfrac{7}{4}$ *cf. 2010 HSC Q6(b)(i)*

**287.** $-\sqrt{3}$

**288.** $x = \dfrac{2\pi}{3}$ or $\dfrac{4\pi}{3}$ *cf. 2011 HSC Q2(b)*

**289.** $x = \dfrac{\pi}{3}$ or $\dfrac{4\pi}{3}$

**290.** (a) $\dfrac{1}{\cos^2 x} + \dfrac{\sin x}{\cos^2 x} = \dfrac{1+\sin x}{\cos^2 x}$ (b) $\cos^2 x = 1 - \sin^2 x = (1-\sin x)(1+\sin x)$, so $\dfrac{1+\sin x}{(1-\sin x)(1+\sin x)} = \dfrac{1}{1-\sin x}$ *cf. 2010 HSC Q5(b)*

**291.** Amplitude 3; period $\dfrac{2\pi}{2} = \pi$. *cf. 2010 HSC Q8(c)*

**292.** $\ell = 6 \times \dfrac{\pi}{3} = 2\pi$ cm

**293.** (a) $\frac{1}{2} \times 9 \times \frac{\pi}{3} = \frac{3\pi}{2}$ cm$^2$ (b) Triangle $AOB$ is equilateral, so $AB = 3$; arc $= 3 \times \frac{\pi}{3} = \pi$. Perimeter $= 3 + \pi$ cm. *cf. 2009 HSC Q5(c)(ii)*

**294.** $2^{2(x-1)} = 2^5$, so $2x - 2 = 5$ and $x = \frac{7}{2}$. *cf. 2011 HSC Q1(c)*

**295.** $\ln 2x = \ln 10$, so $x = 5$.

**296.** $4(x^2 - 3)^3 \times 2x = 8x(x^2 - 3)^3$ *cf. 2008 HSC Q2(a)(i)*

**297.** Tangent gradient 3, normal gradient $-\frac{1}{3}$: $y - 1 = -\frac{1}{3}(x - 1)$, so $x + 3y - 4 = 0$.

**298.** $(x + 5)(x - 1) < 0$, so $-5 < x < 1$. *cf. 2010 HSC Q2(b)*

**299.** $f(-x) = (-x)\sin(-x) = (-x)(-\sin x) = x \sin x = f(x)$ *cf. 2008 HSC Q8(a)(ii)*

**300.** (a) GG, GXG or XGG: $\frac{1}{16} + 2 \times \frac{3}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{16} + \frac{6}{64} = \frac{5}{32}$ (b) The first two games are split: $2 \times \frac{3}{4} \times \frac{1}{4} = \frac{3}{8}$ *cf. 2008 HSC Q7(c)*

**301.** $\frac{6(\sqrt{7} + \sqrt{5})}{7 - 5} = 3(\sqrt{7} + \sqrt{5})$ *cf. 2011 HSC Q1(f)*

**302.** $(x - 3)^2 + (y + 2)^2 = 16$ *cf. 2010 HSC Q1(c)*

**303.** a) $(\cos x)^3$ b) $(\sin 4x)^2$

**304.** $\frac{dy}{dx} = \sin x + x \cos x$

**305.** $\frac{dy}{dx} = 2x \cos x - x^2 \sin x = x(2 \cos x - x \sin x)$

**306.** $\frac{dy}{dx} = 3x^2 \tan x + x^3 \sec^2 x = x^2 \left(3 \tan x + x \sec^2 x\right)$

**307.** $\frac{dy}{dx} = \frac{\sin x}{2\sqrt{x}} + \sqrt{x} \cos x$

**308.** $\frac{dy}{dx} = \sin 3x + 3x \cos 3x$

**309.** $\frac{dy}{dx} = 2x \cos 2x - 2x^2 \sin 2x = 2x(\cos 2x - x \sin 2x)$

**310.** $\frac{dy}{dx} = \cos x \cdot \cos x + \sin x \cdot (-\sin x) = \cos^2 x - \sin^2 x$

**311.** $\frac{dy}{dx} = 2 \cos x \times (-\sin x) = -2 \sin x \cos x$

**312.** $\frac{dy}{dx} = 3 \sin^2 x \cos x$

**313.** $\frac{dy}{dx} = 2 \tan x \sec^2 x$

**314.** $\dfrac{dy}{dx} = \dfrac{-x\sin x - \cos x}{x^2}$

**315.** $\dfrac{dy}{dx} = \dfrac{\sin x - x\cos x}{\sin^2 x}$

**316.** $\dfrac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \dfrac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \dfrac{1+\cos x}{(1+\cos x)^2} = \dfrac{1}{1+\cos x}$

**317.** $\dfrac{dy}{dx} = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x)$

**318.** $\dfrac{dy}{dx} = \dfrac{\cos x}{\sin x}$

**319.** $\dfrac{dy}{dx} = -\sin x\, e^{\cos x}$

**320.** Point $(\pi, -\pi)$. $\dfrac{dy}{dx} = \cos x - x\sin x = -1$. $y + \pi = -(x - \pi)$, so $y = -x$.

**321.** Point $\left(\dfrac{\pi}{4}, \dfrac{1}{2}\right)$. $\dfrac{dy}{dx} = 2\sin x\cos x = 2 \times \dfrac{1}{\sqrt{2}} \times \dfrac{1}{\sqrt{2}} = 1$. $y = x - \dfrac{\pi}{4} + \dfrac{1}{2}$.

**322.** Point $(0, 1)$. $\dfrac{dy}{dx} = e^x(\cos x - \sin x) = 1$, so the normal gradient is $-1$: $y = -x + 1$.

**323.** Quotient rule: $u = \cos x$, $u' = -\sin x$; $v = 1 - \sin x$, $v' = -\cos x$.
$\dfrac{-\sin x(1-\sin x) + \cos^2 x}{(1-\sin x)^2} = \dfrac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \dfrac{1-\sin x}{(1-\sin x)^2} = \dfrac{1}{1-\sin x}$

**324.** $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$ (product rule) and $\dfrac{d}{dx}(\cos x) = -\sin x$. Sum: $\sin x + x\cos x - \sin x = x\cos x$.

**325.** Product rule: $u = e^{-x}$, $u' = -e^{-x}$; $v = \sin x + \cos x$, $v' = \cos x - \sin x$.
$-e^{-x}(\sin x + \cos x) + e^{-x}(\cos x - \sin x) = e^{-x}(-2\sin x) = -2e^{-x}\sin x$

**326.** $2\sin x\cos x + 2\cos x(-\sin x) = 0$. This makes sense because $\sin^2 x + \cos^2 x = 1$, a constant.

**327.** $\dfrac{d}{dx}\ln(\sin x) = \dfrac{\cos x}{\sin x} = \cot x$

**328.** The derivative of a product is not the product of the derivatives. By the product rule, $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$.

**329.** $\dfrac{dy}{dx} = 2x\sin x + x^2\cos x$. At $x = \pi$: $0 + \pi^2(-1) = -\pi^2$.

**330.** a) $f'(x) = -e^{-x}\sin x + e^{-x}\cos x = e^{-x}(\cos x - \sin x)$ b) $e^{-x} > 0$, so $\cos x = \sin x$, $\tan x = 1$: $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

**331.** $2x\cos(x^2+1)$

**332.** $5\sec^2 5x$

**333.** $-3\sin(3x+1)$

**334.** $\dfrac{dy}{dx} = -\sin x = -\dfrac{1}{2}$ *cf. 2010 HSC Q2(c)*

**335.** $2xe^{x^2}$

**336.** $\dfrac{3x^2}{x^3+1}$ *cf. 2011 HSC Q1(d)*

**337.** $3(\ln 2)\, 2^{3x}$

**338.** $\dfrac{1}{x \ln 10}$

**339.** $2xe^x + x^2e^x = xe^x(x+2)$ *cf. 2011 HSC Q2(d)*

**340.** $\dfrac{\frac{1}{x} \times x - \ln x}{x^2} = \dfrac{1 - \ln x}{x^2}$ *cf. 2008 HSC Q2(a)(iii)*

**341.** Point $(1,\, e)$, $m = e$: $y = ex$.

**342.** Point $(1,\, 0)$, tangent gradient 1, normal gradient $-1$: $y = -x + 1$.

**343.** $\cos\dfrac{x}{3} = \pm\dfrac{1}{\sqrt{2}}$ with $-\dfrac{\pi}{3} \le \dfrac{x}{3} \le \dfrac{\pi}{3}$, so $\dfrac{x}{3} = \pm\dfrac{\pi}{4}$ and $x = \pm\dfrac{3\pi}{4}$. *cf. 2008 HSC Q6(a)*

**344.** $2x = 0,\ \pi,\ 2\pi$, so $x = 0,\ \dfrac{\pi}{2},\ \pi$.

**345.** $\cos x = -\dfrac{\sqrt{3}}{2}$: $x = \dfrac{5\pi}{6}$ or $\dfrac{7\pi}{6}$. *cf. 2011 HSC Q2(b)*

**346.** Divide $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $\tan^2 x + 1 = \sec^2 x$. *cf. 2010 HSC Q5(b)*

**347.** $-1 \le y \le 3$

**348.** $\dfrac{\pi}{2}$

**349.** Starts at $(0,\, 1)$, rises to a maximum $(\dfrac{\pi}{2},\, 4)$, returns to $(\pi,\, 1)$. *cf. 2010 HSC Q8(c)(iii)*

**350.** $\ln e^2 + \ln x - \ln x = 2$

**351.** $\log_2 16 = 4$

**352.** $5(1-2x)^4 \times (-2) = -10(1-2x)^4$ *cf. 2008 HSC Q2(a)(i)*

**353.** $\dfrac{2x(x+1) - x^2}{(x+1)^2} = \dfrac{x(x+2)}{(x+1)^2}$

**354.** $x = 1 \pm \sqrt{2}$

**355.** The line has gradient $-\frac{3}{4}$, so the perpendicular gradient is $\frac{4}{3}$. *cf. 2011 HSC Q3(c)(ii)*

**356.** $(a+1)^2 - 1 = a^2 + 2a$

**357.** RYR: $\frac{2}{5}\times\frac{3}{4}\times\frac{1}{3}=\frac{1}{10}$; YRY: $\frac{3}{5}\times\frac{2}{4}\times\frac{2}{3}=\frac{1}{5}$. Total $\frac{3}{10}$. *cf. 2011 HSC Q5(b)(iii)*

**358.** $5\sqrt{2}-3\sqrt{2}=2\sqrt{2}$

**359.** a) $(\sin 5x)^{-1}$ b) $\left(\cos\left(x^3\right)\right)^{-1}$

**360.** a) $\frac{2}{\sqrt{3}}$ b) $\frac{2}{\sqrt{3}}$ c) 1 d) 1

**361.** $\frac{d}{dx}(\cos 3x)^{-1}=-(\cos 3x)^{-2}\times(-3\sin 3x)=\frac{3\sin 3x}{\cos^2 3x}=3\tan 3x\sec 3x$

**362.** $y=(\cos x)^{-2}$, so $\frac{dy}{dx}=-2(\cos x)^{-3}(-\sin x)=\frac{2\sin x}{\cos^3 x}=2\tan x\sec^2 x.$

**363.** $\frac{dy}{dx}=5\tan x\sec x$

**364.** $\frac{dy}{dx}=-\cot x\operatorname{cosec} x+2x$

**365.** $\frac{dy}{dx}=-2\cot x\operatorname{cosec} x$

**366.** $\frac{d}{dx}(\sin 2x)^{-1}=-(\sin 2x)^{-2}\times 2\cos 2x=-2\cdot\frac{\cos 2x}{\sin 2x}\cdot\frac{1}{\sin 2x}=-2\cot 2x\operatorname{cosec} 2x$

**367.** $\frac{dy}{dx}=-\operatorname{cosec}^2 x-1$

**368.** $\frac{dy}{dx}=-4\operatorname{cosec}^2 x$

**369.** $\frac{d}{dx}(\tan x)^{-1}=-(\tan x)^{-2}\sec^2 x=-\frac{\cos^2 x}{\sin^2 x}\cdot\frac{1}{\cos^2 x}=-\frac{1}{\sin^2 x}=-\operatorname{cosec}^2 x$

**370.** $\frac{dy}{dx}=\tan x\sec x-\cot x\operatorname{cosec} x$

**371.** $\frac{dy}{dx}=-2\operatorname{cosec}^2 x-3\tan x\sec x$

**372.** $\frac{dy}{dx}=\sec^2 x-\operatorname{cosec}^2 x$

**373.** $\frac{dy}{dx}=2x\tan\left(x^2\right)\sec\left(x^2\right)$

**374.** $\frac{dy}{dx}=-3\operatorname{cosec}^2 3x$

**375.** $\frac{dy}{dx}=-2\cot(2x-1)\operatorname{cosec}(2x-1)$

**376.** $\frac{dy}{dx}=\operatorname{cosec} x-x\cot x\operatorname{cosec} x=\operatorname{cosec} x(1-x\cot x)$

**377.** $\dfrac{dy}{dx} = 2x\cot x - x^2\operatorname{cosec}^2 x = x\left(2\cot x - x\operatorname{cosec}^2 x\right)$

**378.** $\dfrac{dy}{dx} = e^x\sec x + e^x\tan x\sec x = e^x\sec x(1+\tan x)$

**379.** Point $\left(\dfrac{\pi}{3}, 2\right)$. $m = \tan\dfrac{\pi}{3}\sec\dfrac{\pi}{3} = \sqrt{3}\times 2 = 2\sqrt{3}$. $y - 2 = 2\sqrt{3}\left(x - \dfrac{\pi}{3}\right)$, so $y = 2\sqrt{3}x - \dfrac{2\sqrt{3}\pi}{3} + 2$.

**380.** Tangent gradient $-\cot\dfrac{\pi}{4}\operatorname{cosec}\dfrac{\pi}{4} = -\sqrt{2}$, so the normal gradient is $\dfrac{1}{\sqrt{2}}$.

**381.** Point $\left(\dfrac{\pi}{2}, 0\right)$. $m = -\operatorname{cosec}^2\dfrac{\pi}{2} = -1$. The tangent is $y = -x + \dfrac{\pi}{2}$.

**382.** $\sec x = (\cos x)^{-1}$, so $\dfrac{d}{dx}(\sec x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos x} = \tan x\sec x$.

**383.** $\cot x = \dfrac{\cos x}{\sin x}$, so $\dfrac{d}{dx}(\cot x) = \dfrac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$.

**384.** $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x = -\dfrac{1}{\sin^2 x}$. Since $\sin^2 x > 0$ wherever $\cot x$ is defined, the gradient is always negative.

**385.** The derivative of a reciprocal is not the reciprocal of the derivative. Using the chain rule on $(\cos x)^{-1}$ gives $\dfrac{\sin x}{\cos^2 x} = \tan x\sec x$.

**386.** a) $f'(x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos^2 x}$ b) $\sin x = 0$ gives $x = 0$; the point is $(0, 1)$. c) For $0 < x < \dfrac{\pi}{2}$, $\sin x > 0$ and $\cos^2 x > 0$.

**387.** $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$ *cf. 2009 HSC Q2(a)(i)*

**388.** $2\cos x\times(-\sin x) = -2\sin x\cos x$

**389.** $\tan 2x + 2x\sec^2 2x$ *cf. 2010 HSC Q1(e)*

**390.** $\cos x\, e^{\sin x}$

**391.** $\dfrac{2x\cos 2x - \sin 2x}{x^2}$ *cf. 2010 HSC Q2(a)*

**392.** $\dfrac{\cos x}{\sin x} = \cot x$ *cf. 2008 HSC Q3(b)(i)*

**393.** $\dfrac{-(x+2)\sin x - \cos x}{(x+2)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**394.** $2xe^{-x} - x^2e^{-x} = xe^{-x}(2-x)$ *cf. 2011 HSC Q2(d)*

**395.** Point $(\pi, 0)$, $m = \cos\pi = -1$: $y = -x + \pi$.

**396.** $\sec^2\dfrac{\pi}{3} = 2^2 = 4$ *cf. 2010 HSC Q2(c)*

**397.** $\tan x = 1$: $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

**398.** $\dfrac{2x\cos x + x^2 \sin x}{\cos^2 x}$ *cf. 2011 HSC Q4(a)*

**399.** $-\dfrac{\sqrt{3}}{2}$

**400.** $\sin^2 x \times \dfrac{1}{\sin x} = \sin x$

**401.** $\operatorname{cosec}^2 x$

**402.** $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$ *cf. 2011 HSC Q2(b)*

**403.** $x - 2 = 5$, so $x = 7$.

**404.** $e^{2x} = 4$, so $x = \frac{1}{2}\ln 4 = \ln 2$.

**405.** $\log_3 1 = 0$

**406.** $12x^3 + \dfrac{2}{x^2}$

**407.** $\dfrac{x}{\sqrt{x^2+4}}$ *cf. 2011 HSC Q4(d)(i)*

**408.** $\dfrac{dy}{dx} = 3x^2 + 3 \geq 3 > 0$ for all $x$, so the gradient is never 0. *cf. 2010 HSC Q6(a)(i)*

**409.** $x < -3$ or $x > 3$ *cf. 2010 HSC Q2(b)*

**410.** $x^2 - 2x + 3 - c = 0$ has equal roots: $4 - 4(3 - c) = 0$, so $c = 2$. *cf. 2009 HSC Q4(b)*

**411.** $x^2 - x - 2 = 0$, so $x = 2$ or $-1$: $(2, 4)$ and $(-1, 1)$.

**412.** $x^2 > 0$, so all real $x$ except $x = 0$. *cf. 2010 HSC Q1(g)*

**413.** (a) $\dfrac{1}{4}$ (b) The car is on the last level he would search: $\dfrac{3}{4} \times \dfrac{2}{3} \times \dfrac{1}{2} = \dfrac{1}{4}$ (c) $\left(\dfrac{3}{4}\right)^5 = \dfrac{243}{1024}$ *cf. 2009 HSC Q5(b)*

**414.** $27a^6b^3$

**415.** a) $e^x$ b) $\dfrac{1}{x}$ c) $\cos x$ d) $-\operatorname{cosec}^2 x$ e) $(\ln 3)\, 3^x$

**416.** a) Product b) Chain c) Quotient d) Term by term

**417.** $\dfrac{dy}{dx} = e^x \cos 2x - 2e^x \sin 2x = e^x(\cos 2x - 2\sin 2x)$

**418.** $\dfrac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$

**419.** $\dfrac{dy}{dx} = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$

**420.** $\dfrac{dy}{dx} = e^{-3x} - 3xe^{-3x} = e^{-3x}(1-3x)$

**421.** $\dfrac{dy}{dx} = -e^{-x}\cos x - e^{-x}\sin x = -e^{-x}(\cos x + \sin x)$

**422.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}e^x - e^x \ln x}{e^{2x}} = \dfrac{\frac{1}{x} - \ln x}{e^x} = \dfrac{1 - x\ln x}{xe^x}$

**423.** $\dfrac{dy}{dx} = \dfrac{e^{2x}\cos x - 2e^{2x}\sin x}{e^{4x}} = \dfrac{\cos x - 2\sin x}{e^{2x}}$

**424.** $\dfrac{dy}{dx} = \dfrac{2x\ln x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{x(2\ln x - 1)}{(\ln x)^2}$

**425.** $\dfrac{dy}{dx} = -e^x \sin(e^x)$

**426.** $\dfrac{dy}{dx} = \sec^2 x\, e^{\tan x}$

**427.** $\dfrac{dy}{dx} = \dfrac{e^x}{1+e^x}$

**428.** $\dfrac{dy}{dx} = \dfrac{1-\sin x}{x+\cos x}$

**429.** $\dfrac{dy}{dx} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

**430.** $\dfrac{dy}{dx} = \dfrac{2 + 2\cos 2x}{2x + \sin 2x}$

**431.** $\dfrac{dy}{dx} = 3(e^x+1)^2 \times e^x = 3e^x(e^x+1)^2$

**432.** $\dfrac{dy}{dx} = 2(\sin x + \cos x)(\cos x - \sin x)$

**433.** $y = (1+\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(1+\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{1+\ln x}}$.

**434.** $\dfrac{\cos x(1+\cos x)^2 + 2\sin^2 x(1+\cos x)}{(1+\cos x)^4} = \dfrac{\cos x + \cos^2 x + 2 - 2\cos^2 x}{(1+\cos x)^3} = \dfrac{(2-\cos x)(1+\cos x)}{(1+\cos x)^3} = \dfrac{2-\cos x}{(1+\cos x)^2}$

**435.** $\dfrac{dy}{dx} = \dfrac{(x^2+1)^2 - x\cdot 4x(x^2+1)}{(x^2+1)^4} = \dfrac{x^2+1-4x^2}{(x^2+1)^3} = \dfrac{1-3x^2}{(x^2+1)^3}$

**436.** $\dfrac{dy}{dx} = \dfrac{e^x(e^x+1)^3 - e^x \cdot 3e^x(e^x+1)^2}{(e^x+1)^6} = \dfrac{e^x(1-2e^x)}{(e^x+1)^4}$

**437.** $\dfrac{dy}{dx} = e^x(\sin x + \cos x) = 1$ at $x = 0$. The tangent is $y = x$.

**438.** Point $(1, \ln 2)$. $\dfrac{dy}{dx} = \dfrac{2x}{1+x^2} = 1$. $y = x - 1 + \ln 2$.

**439.** $\dfrac{dy}{dx} = e^{2x}(1+2x) = 1$ at $x = 0$, so the normal gradient is $-1$: $y = -x$.

**440.** In $e^x \sin x$ the last operation is multiplying two functions; in $e^{\sin x}$ one function is inside another. $\dfrac{d}{dx}e^x \sin x = e^x(\sin x + \cos x)$ and $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$.

**441.** $\dfrac{d}{dx}\ln(\sec x) = \dfrac{\tan x \sec x}{\sec x} = \tan x$

**442.** The student used the power rule on an exponential. For $e^{f(x)}$ the power stays the same and is multiplied by its derivative: $2xe^{x^2}$.

**443.** $\dfrac{dy}{dx} = \ln x + 1 - 1 = \ln x$, so the gradient at $x = e$ is 1.

**444.** a) $f'(x) = \dfrac{\frac{1}{x}\cdot x^2 - 2x\ln x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$ b) $\ln x = \frac{1}{2}$, so $x = \sqrt{e}$ and $f(\sqrt{e}) = \dfrac{\frac{1}{2}}{e} = \dfrac{1}{2e}$. The point is $\left(\sqrt{e}, \dfrac{1}{2e}\right)$.

**445.** $5e^{5x-1}$

**446.** $\ln 3 + 2\ln x$, so $\dfrac{2}{x}$. *cf. 2010 HSC Q2(c)*

**447.** $\dfrac{1}{(x+1)\ln 2}$

**448.** $(\ln 7)\, 7^x$

**449.** $2x\sec^2(x^2)$

**450.** $3\tan 3x \sec 3x$

**451.** $-\cot x \operatorname{cosec} x$

**452.** $-2\operatorname{cosec}^2 2x$

**453.** $2\sin 3x \times 3\cos 3x = 6\sin 3x \cos 3x$

**454.** $\cos x - x\sin x$ *cf. 2009 HSC Q2(a)(i)*

**455.** Point $(0, 1)$, $m = -1$: $y = -x + 1$.

**456.** Tangent gradient $\cos 0 = 1$, normal gradient $-1$: $y = -x$.

**457.** $\dfrac{e^x(e^x+1) - (e^x-1)e^x}{(e^x+1)^2} = \dfrac{2e^x}{(e^x+1)^2}$ *cf. 2010 HSC Q4(d)*

**458.** (a) $g'(x) = \dfrac{1}{1+x}$ (b) For $x \geq 0$, $1 + x \geq 1$, so $\dfrac{1}{1+x} \leq 1$. *cf. 2009 HSC Q10(c)(ii)*

**459.** $\dfrac{1}{\cos\frac{\pi}{4}} = \sqrt{2}$

**460.** $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$

**461.** $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$, so $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

**462.** $x = \ln 3$

**463.** $x = 25$

**464.** $\ln e^2 = 2$

**465.** $-6(3x+2)^{-3}$

**466.** $\dfrac{2x(1-x)+x^2}{(1-x)^2} = \dfrac{x(2-x)}{(1-x)^2}$ *cf. 2008 HSC Q2(a)(iii)*

**467.** Point $(-1, -2)$, $m = 2(-1)+3 = 1$: $y = x-1$. *cf. 2011 HSC Q2(c)*

**468.** $(2x-1)(x-3) = 0$: $x = \frac{1}{2}$ or 3.

**469.** (a) $x^2(x^2-2) = 0$: $(0, 0)$, $(\sqrt{2}, 0)$, $(-\sqrt{2}, 0)$; $y$-intercept $(0, 0)$. (b) $f(-x) = x^4 - 2x^2 = f(x)$ *cf. 2008 HSC Q8(a)*

**470.** $3x-1 = 5$ or $3x-1 = -5$: $x = 2$ or $x = -\frac{4}{3}$. *cf. 2010 HSC Q1(d)*

**471.** $x \geq 3$ *cf. 2010 HSC Q1(g)*

**472.** $p^2 = 0.49$, so $P(\text{head}) = 0.7$ and $P(\text{both tails}) = 0.3^2 = 0.09$. *cf. 2010 HSC Q8(b)*

**473.** (a) $\dfrac{dy}{dx} = 2x = -2$: $y - 1 = -2(x+1)$, so $y = -2x - 1$. (b) $AB$ has gradient $\dfrac{4-1}{2+1} = 1$; $2x = 1$, so $C\left(\frac{1}{2}, \frac{1}{4}\right)$. *cf. 2010 HSC Q7(b)*

**474.** $x^{-6} \times x^7 = x$